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PE A 3x3 grid of lines, consisting of three horizontal and three vertical lines intersecting to form a 3x3 grid of squares.

civil: structural
practice exam

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About NCEES

NCEES is a nonprofit organization made up of the U.S. engineering and surveying licensing boards in all 50 states, the U.S. territories, and the District of Columbia. We develop and score the exams used for engineering and surveying licensure in the United States. NCEES also promotes professional mobility through its services for licensees and its member boards.

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Exam format

The PE Civil: Structural exam is computer-based. It contains 80 questions and is administered year-round via computer at approved Pearson VUE test centers. A 9-hour appointment time includes a tutorial, the exam, and a break. You have 8 hours to complete the actual exam.

In addition to traditional multiple-choice questions with one correct answer, the exam uses common alternative item types such as

- Multiple correct options—allows multiple choices to be correct
- Point and click—requires examinees to click on part of a graphic to answer
- Drag and drop—requires examinees to click on and drag items to match, sort, rank, or label
- Fill in the blank—provides a space for examinees to enter a response to the question

To familiarize yourself with the format, style, and navigation of a computer-based exam, view the video tutorials on the NCEES YouTube channel.

Examinee Guide

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Scoring and reporting

Results for computer-based exams are typically available 7–10 days after you take the exam. You will receive an email notification from NCEES with instructions to view your results in your MyNCEES account. All results are reported as pass or fail.

Updates on exam content and procedures

Visit us at ncees.org/exams for updates on everything exam-related, including specifications, exam-day policies, scoring, and corrections to published exam preparation materials. This is also where you will register for the exam and find additional steps you should follow in your state to be approved for the exam.



PE CIVIL: STRUCTURAL PRACTICE EXAM

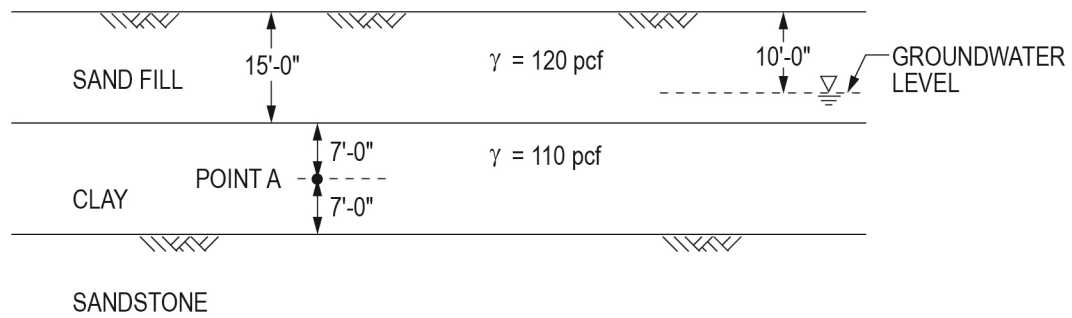
PE CIVIL: STRUCTURAL PRACTICE EXAM

1. A bridge is to be jacked up to replace its bearings. The design requires a hydraulic ram with a minimum capacity of 1,000 kilonewtons (kN). The hydraulic rams that are available are rated in tons (2,000 lb/ton). The **minimum** size (tons) ram to use is most nearly:
 - ☐ A. 1,110
 - ☐ B. 250
 - ☐ C. 150
 - ☐ D. 100

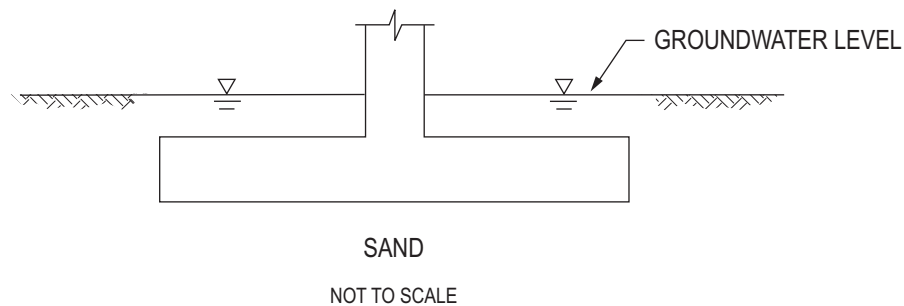
2. Which one of the following statements regarding lateral earth pressures is correct?
 - ☐ A. The lateral strain required to fully mobilize the soil passive pressure is considerably smaller than the lateral strain required to fully mobilize the soil active pressure.
 - ☐ B. The lateral strain required to fully mobilize the soil passive pressure is slightly smaller than the lateral strain required to fully mobilize the soil active pressure.
 - ☐ C. The lateral strain required to fully mobilize the soil passive pressure is slightly larger than the lateral strain required to fully mobilize the soil active pressure.
 - ☐ D. The lateral strain required to fully mobilize the soil passive pressure is considerably larger than the lateral strain required to fully mobilize the soil active pressure.

PE CIVIL: STRUCTURAL PRACTICE EXAM

3. A soil profile is shown. The effective vertical stress (psf) at Point A is most nearly:



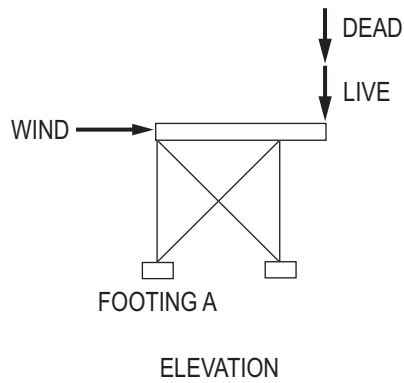
- ☐ A. 1,270
 - ☐ B. 1,820
 - ☐ C. 2,140
 - ☐ D. 2,570
4. A bridge footing is to be constructed in sand. The groundwater level is at the ground surface. The ultimate bearing capacity is based on what type of soil unit weight?



- ☐ A. Buoyant unit weight
- ☐ B. Saturated unit weight
- ☐ C. Dry unit weight
- ☐ D. Total unit weight

PE CIVIL: STRUCTURAL PRACTICE EXAM

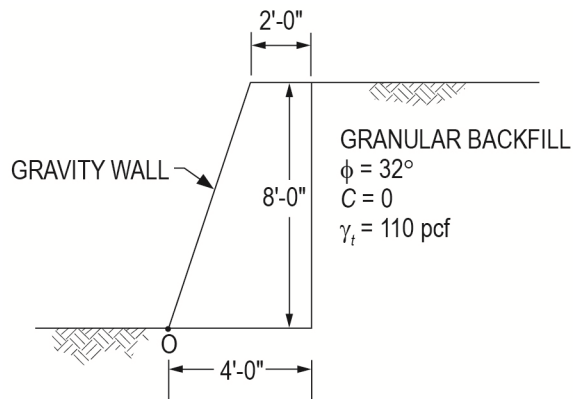
5. In the figure, what load combination produces the maximum uplift on Footing A?



- ☐ A. Dead
- ☐ B. Dead + live
- ☐ C. Dead + wind
- ☐ D. Dead + live + wind

PE CIVIL: STRUCTURAL PRACTICE EXAM

6. The figure shows a concrete gravity retaining wall having a unit weight of 150 pcf. Use the Rankine active earth pressure theory and neglect wall friction. The factor of safety against overturning about the toe at Point O is most nearly:



- ☐ A. 3.0
- ☐ B. 2.5
- ☐ C. 2.2
- ☐ D. 0.3

PE CIVIL: STRUCTURAL PRACTICE EXAM

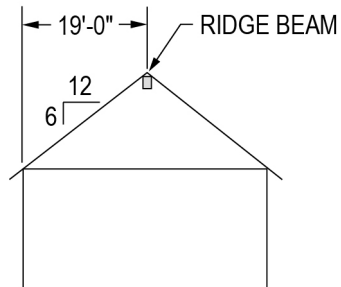
7. The following data apply to the structure shown in the figure:

Ground snow load = 20 psf
Roof is fully exposed with wood shingles
Open terrain
Risk Category I
Unheated structure

Design Code:

ASCE 7-16: *Minimum Design Loads for Buildings and Other Structures*.

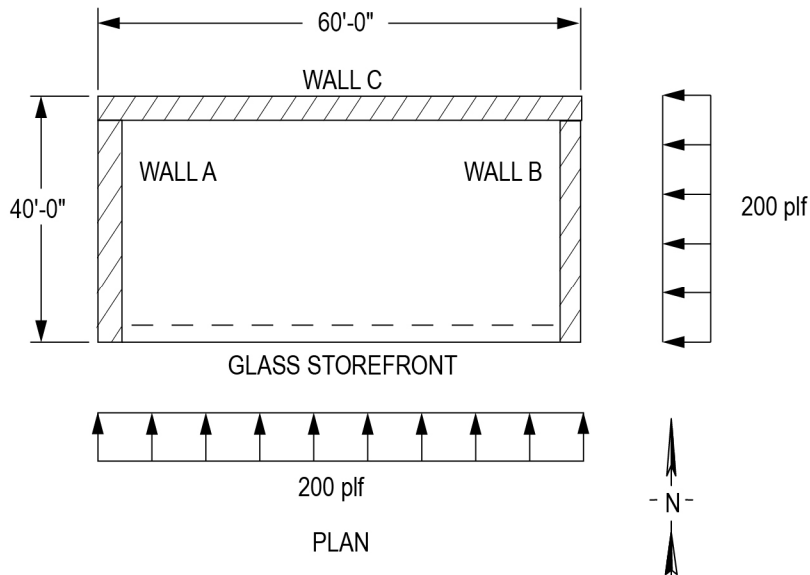
Roof joists span from the exterior walls to the ridge beam. The design snow load (psf) for roof joists is most nearly:



- ☐ A. 10
- ☐ B. 12
- ☐ C. 16
- ☐ D. 20

PE CIVIL: STRUCTURAL PRACTICE EXAM

8. The figure shows a one-story building with CMU walls on three sides and a glass storefront. The uniform load shown in the figure is the wind reaction acting at the top of the wall, directly into the diaphragm. The resultant force (kips) on Wall B from the wind acting on the front wall is most nearly:



- ☐ A. 4
- ☐ B. 6
- ☐ C. 8
- ☐ D. 12

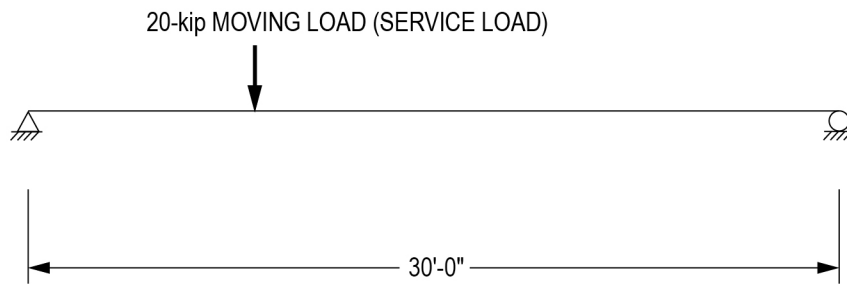
PE CIVIL: STRUCTURAL PRACTICE EXAM

9. A two-story 100-ft $\times$ 100-ft industrial facility in Seismic Design Category A has a second floor that is used for storage and a 200-psf reduced floor live load. The facility has a flat roof that experiences a snow load of 50 psf, and the total dead load of the building is 200,000 lb. According to ASCE 7-16, the minimum lateral force E (lb) for the entire building is most nearly:

- ☐ A. 2,000
- ☐ B. 7,000
- ☐ C. 22,000
- ☐ D. 27,000

10. A moving load is applied to a beam as shown in the figure. The maximum service moment (ft-kips) carried by the beam is _____.

Enter your response in the blank.



PE CIVIL: STRUCTURAL PRACTICE EXAM

11. A building floor is supported by steel beams as shown in the figure. The floor beams are spaced 10 ft o.c. at either side of the beam shown.

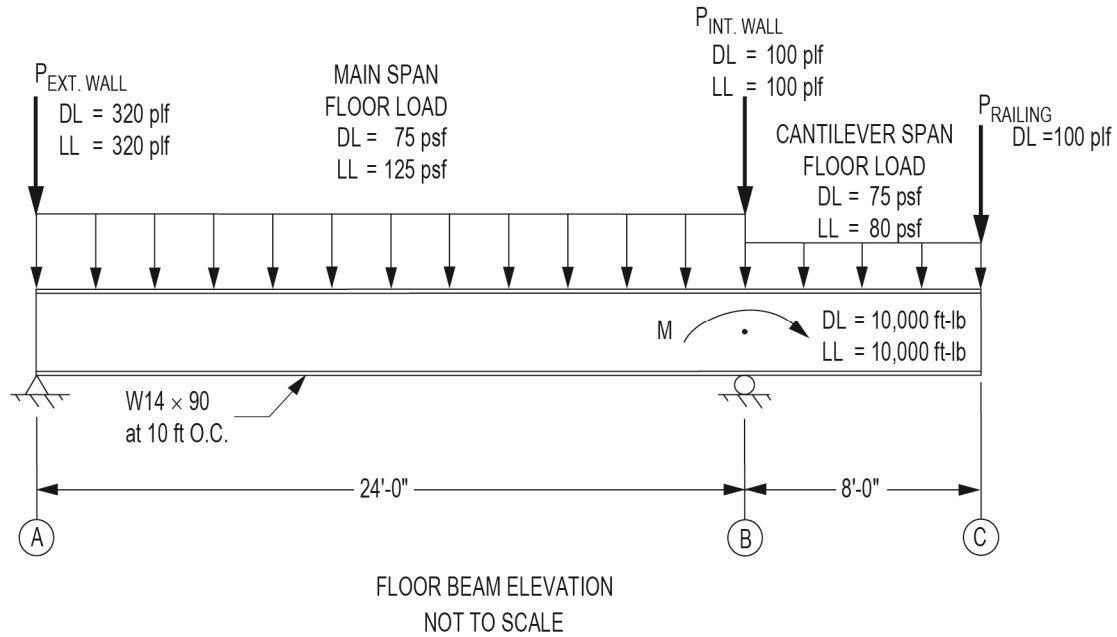
Assumptions:

All loads act simultaneously.

Neglect beam self-weight.

The live loads are nonreducible.

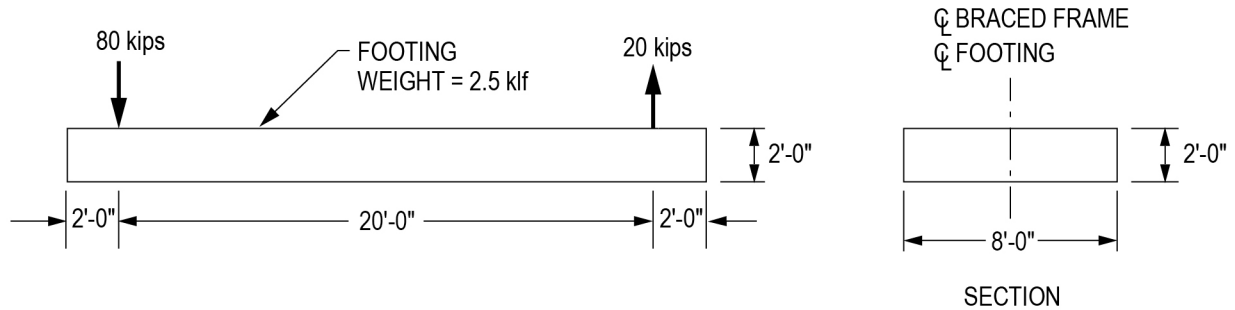
The total service load reaction (kips) at the support at A is most nearly:



- ☐ A. 3
- ☐ B. 14
- ☐ C. 28
- ☐ D. 35

PE CIVIL: STRUCTURAL PRACTICE EXAM

12. The concrete footing shown is subjected to loads from a braced frame. Assume the footing is rigid compared to the soil. The maximum bearing pressure (ksf) under the footing is most nearly:



- A. 1.61
- B. 1.93
- C. 2.72
- D. 4.17
13. A building is to be constructed on a site defined as Site Class D with a 0.2-sec mapped spectral response acceleration of 100% g. Per IBC 2018, the design spectral response acceleration at short periods is most nearly:
- A. 0.60
- B. 0.73
- C. 0.90
- D. 1.60

PE CIVIL: STRUCTURAL PRACTICE EXAM

14. A building is located in a region with a design wind speed of 120 mph. The rectangular building is 80 ft by 120 ft and has a flat roof. According to ASCE 7-16, the wind velocity pressure (psf) on the roof of the building is _____.

Assumptions:

Risk category is III.

$$K_h = 0.9$$

$$K_{zt} = 1.0$$

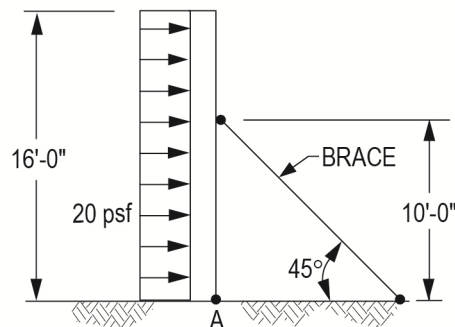
$$K_d = 0.85$$

$$K_e = 1.0$$

$$K_f = 1.3$$

Enter your response in the blank.

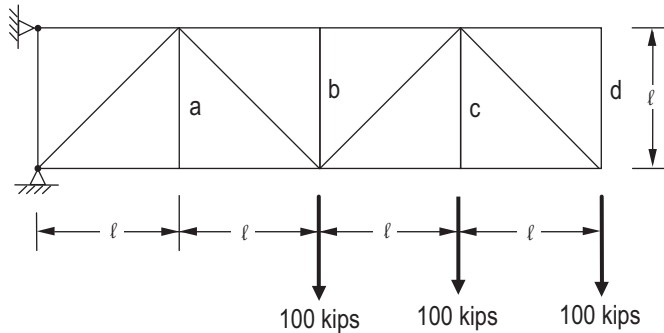
15. A wall form subjected to a wind load of 20 psf is prevented from overturning by diagonal braces spaced at 8 ft o.c. along the length of the wall form as shown in the figure. The connection at the base of the form at Point A is equivalent to a hinge. Ignore the weight of the form. The axial force (lb) resisted by the brace is most nearly:



- ☐ A. 2,050
- ☐ B. 2,560
- ☐ C. 2,900
- ☐ D. 4,525

PE CIVIL: STRUCTURAL PRACTICE EXAM

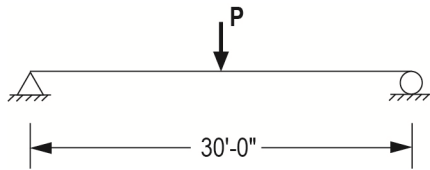
16. A simply supported truss is loaded as shown in the figure. The loads (kips) for Members b and c are most nearly:



- ☐ A. Member b = 0
Member c = 0
 - ☐ B. Member b = 0
Member c = 100
 - ☐ C. Member b = 100
Member c = 0
 - ☐ D. Member b = 100
Member c = 100
17. Consider two beams with equal cross sections, made of the same material, having the same support conditions, and each loaded with equal uniform load per length. One beam is twice as long as the other. The maximum bending stress in the longer beam is larger by a factor of:
- ☐ A. 1.25
 - ☐ B. 2
 - ☐ C. 3
 - ☐ D. 4

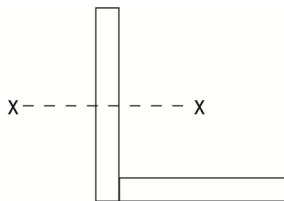
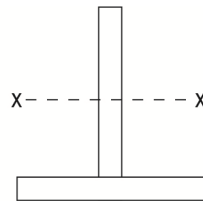
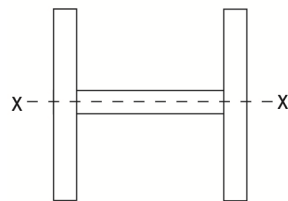
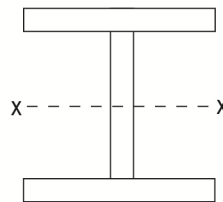
PE CIVIL: STRUCTURAL PRACTICE EXAM

18. The point load (kips) placed at the centerline of a 30-ft beam that produces the same maximum shear in the beam as a uniform load of 1 kip/ft is most nearly:



- ☐ A. 7.5
- ☐ B. 15
- ☐ C. 30
- ☐ D. 60

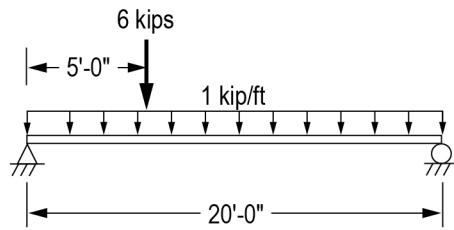
19. The beam sections shown are fabricated from 1/2-in. $\times$ 6-in. steel plates. Which of the following cross sections will provide the greatest flexural rigidity about the x-axis?

**Option A****Option B****Option C****Option D**

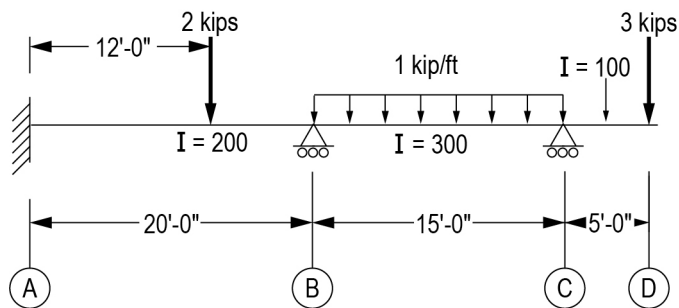
- ☐ A. Option A
- ☐ B. Option B
- ☐ C. Option C
- ☐ D. Option D

PE CIVIL: STRUCTURAL PRACTICE EXAM

20. The beam shown in the figure is loaded with a uniform load of 1 kip/ft and a point load of 6 kips located 5 ft from the left end. The location of the maximum moment (ft) measured from the right end is most nearly:



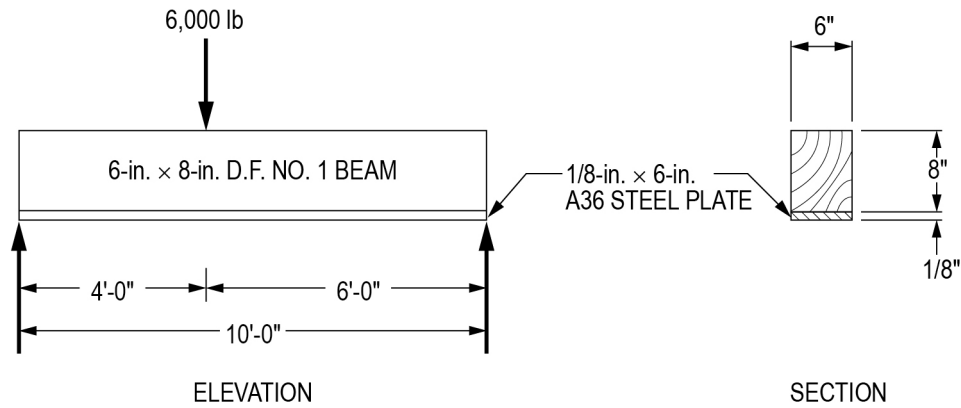
- A. 5.0
 ○ B. 8.5
 ○ C. 10.0
 ○ D. 11.5
21. A loaded beam is shown in the figure. Using the method of moment distribution, the unbalanced portion of the fixed-end moments (ft-kips) at Joint B is most nearly:



- A. 19
 ○ B. 13
 ○ C. 6
 ○ D. 0

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22. The beam shown in the figure is simply supported, and the dead load is negligible. A steel plate is attached to the beam sufficiently to fully develop composite section. $E_{\text{wood}} = 1.6 \times 10^6$ psi and $E_{\text{steel}} = 29 \times 10^6$ psi. The tensile stress in the steel (ksi) is most nearly:



- ☐ A. 2.6
- ☐ B. 14.4
- ☐ C. 18.6
- ☐ D. 23.5

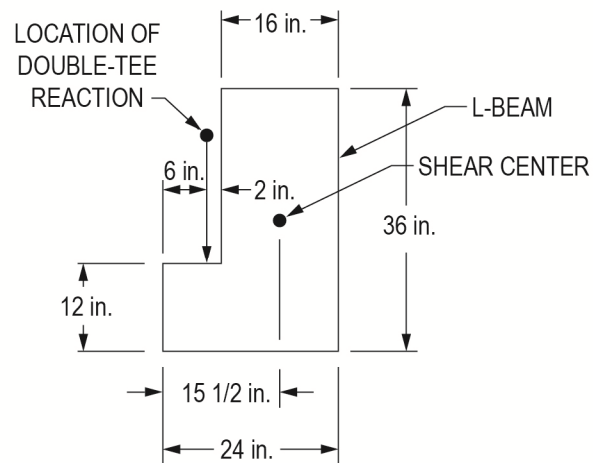
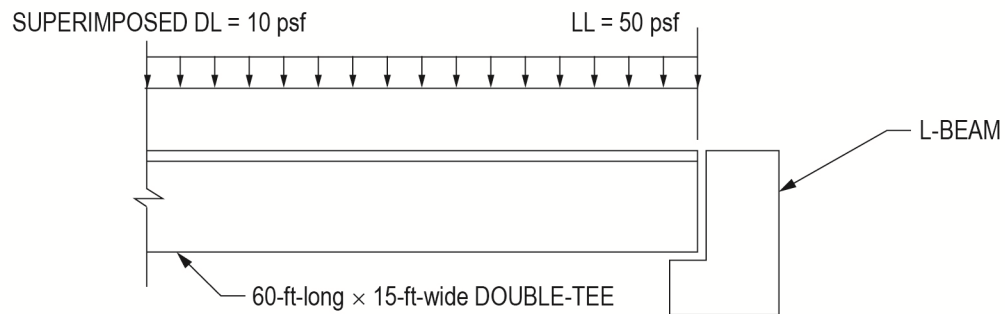
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23. A double-tee supported by an L-beam is loaded as shown in the figure.

Design Code:

ACI 318: *Building Code Requirements for Structural Concrete*, 2014

The magnitude of the ultimate torsion (ft-kips) induced into the L-beam, by superimposed dead and live load applied on the double-tee, is most nearly:

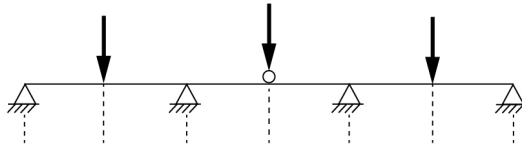


NOT TO SCALE

- ☐ A. 18.7
- ☐ B. 21.4
- ☐ C. 28.6
- ☐ D. 32.8

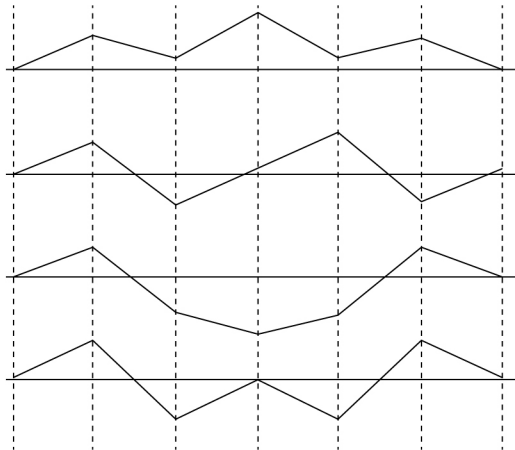
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24. A beam is loaded as shown:

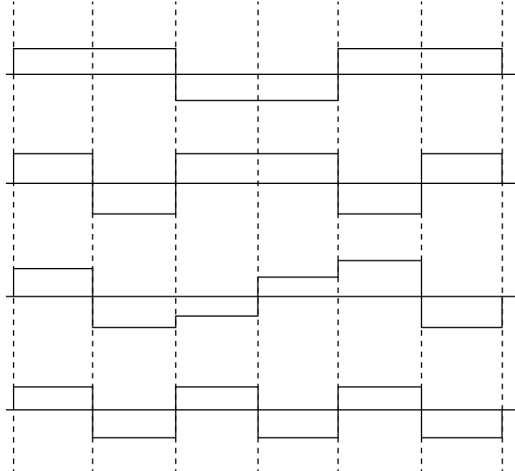


Select the two diagrams that most accurately represent the shear and bending moment for the loaded beam shown.

MOMENT DIAGRAMS

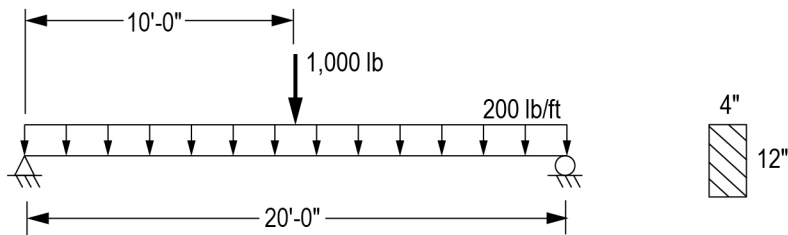


SHEAR DIAGRAMS



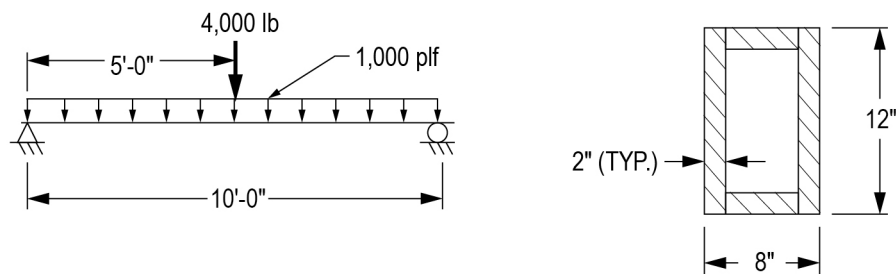
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25. Referring to the figure, the maximum flexural stress (psi) in the member is most nearly:



- ☐ A. 5,625
- ☐ B. 3,125
- ☐ C. 1,875
- ☐ D. 1,565

26. The figure shows a section with elements that are adequately connected to ensure composite action. Assuming $Q = 85 \text{ in}^3$, the maximum horizontal shear stress (psi) in the member is most nearly:



- ☐ A. 90
- ☐ B. 150
- ☐ C. 300
- ☐ D. 400

PE CIVIL: STRUCTURAL PRACTICE EXAM

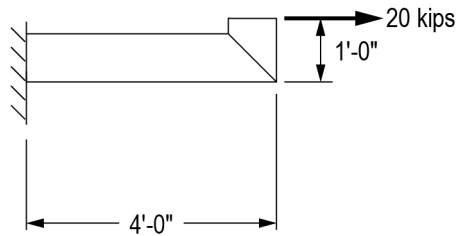
27. The pipe member shown in the figure has a constant section and the following properties:

Outside diameter = 10 in.

Moment of inertia = 90 in^4

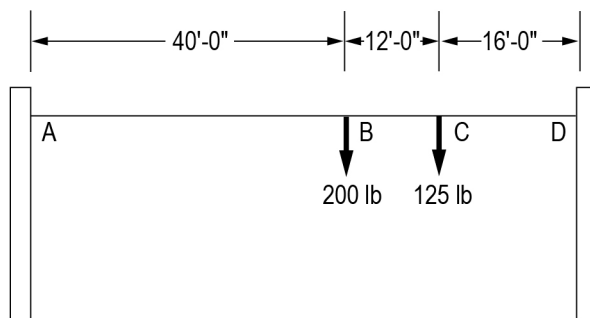
Area = 7 in^2

Neglecting the weight of the pipe, the maximum compressive stress (ksi) at the support is most nearly:



- ☐ A. 2.9
- ☐ B. 4.9
- ☐ C. 13.3
- ☐ D. 16.2

28. A cable carrying traffic signal loads is shown in the figure. The deflection at Point B is 12 in. Neglect elongation of the cable. The tension force (lb) in the cable between C and D is most nearly:



- ☐ A. 4,685
- ☐ B. 4,476
- ☐ C. 1,816
- ☐ D. 213

PE CIVIL: STRUCTURAL PRACTICE EXAM

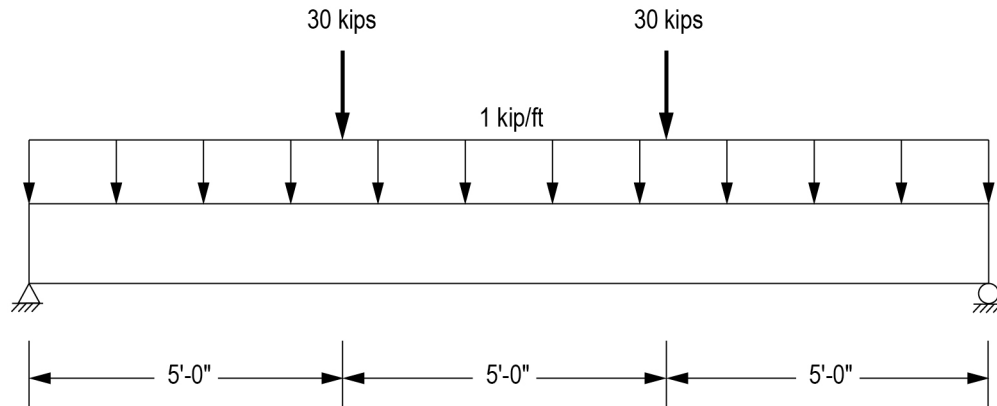
29. The following information applies to the simply supported steel beam shown in the figure:

$$E = 29,000 \text{ ksi}$$

Uniform load includes beam self-weight

Considering only deflection criteria, which W sections meet a maximum deflection criteria of $\ell/240$?

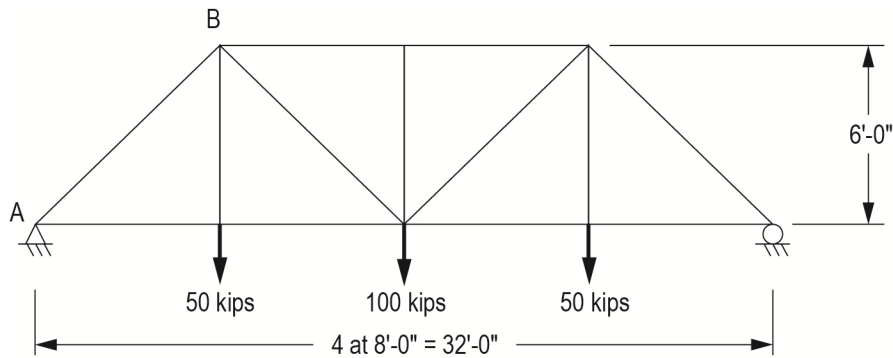
Select the **four** that apply.



- ☐ A. W12×40
- ☐ B. W12×45
- ☐ C. W12×50
- ☐ D. W14×26
- ☐ E. W14×30
- ☐ F. W14×34
- ☐ G. W16×26
- ☐ H. W16×31

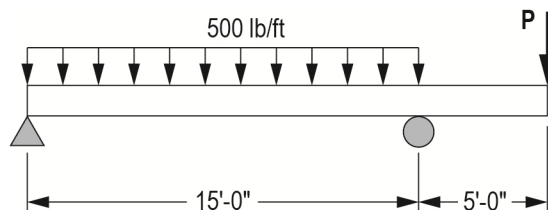
PE CIVIL: STRUCTURAL PRACTICE EXAM

30. Referring to the loaded truss shown in the figure, the force (kips) in Member AB is most nearly:



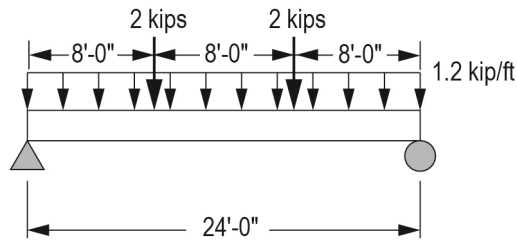
- A. 100 (compression)
 ○ B. 125 (compression)
 ○ C. 167 (tension)
 ○ D. 167 (compression)
31. An A992 W8×28 beam is loaded as shown. To have zero deflection at the free end of the overhanging section, the magnitude (kips) of the concentrated force, **P**, at the free end must be _____.

Enter your response in the blank.



PE CIVIL: STRUCTURAL PRACTICE EXAM

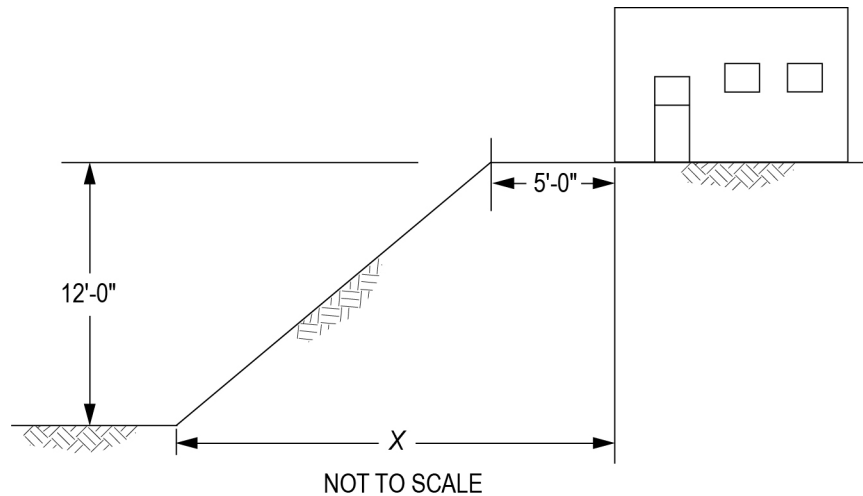
32. An A992 W12×53 beam is loaded as shown. The maximum deflection (in.) of this beam is most nearly:



- A. 0.138
- B. 0.727
- C. 0.864
- D. 3.835
33. A 20-ft-long A992 W14×53 beam is pinned at both ends. It is subjected to an increase in temperature of 200°F. Assume the temperature coefficient of expansion for steel is $6.5 \times 10^{-6}/^{\circ}\text{F}$. The axial stress (ksi) due to this increase in temperature is most nearly:
- A. 18.9
- B. 26.0
- C. 37.7
- D. 60.0

PE CIVIL: STRUCTURAL PRACTICE EXAM

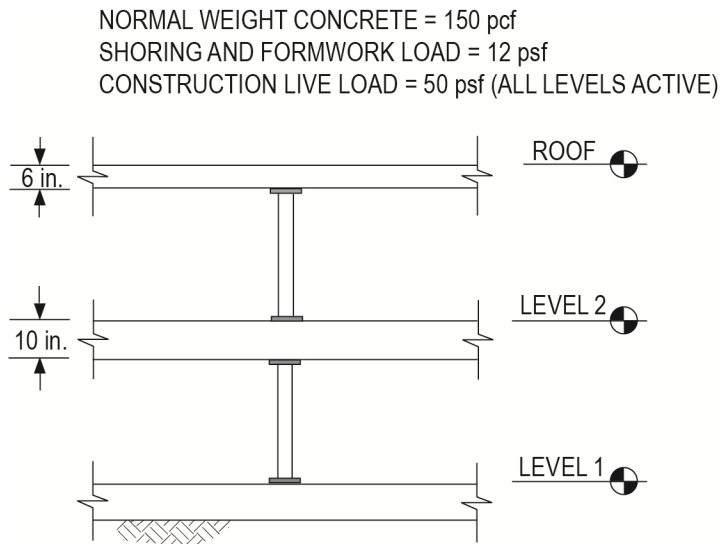
34. Based on the soil classification system found in the federal OSHA excavation regulations, the soil adjacent to an existing building has been classified as Type B. An undisturbed perimeter strip that is 5 ft wide is to be maintained along the face of the building. The excavation is to be 12 ft deep. To meet OSHA excavation requirements, the minimum horizontal distance X (ft) from the toe of the slope to the face of the structure is most nearly:



- ☐ A. 11
- ☐ B. 14
- ☐ C. 17
- ☐ D. 23

PE CIVIL: STRUCTURAL PRACTICE EXAM

35. The formwork and shoring are in place at both levels of the two-story building shown. Shores are spaced at 6 ft o.c. Assume all Level 2 loads are carried by the formwork and shoring during the roof pour. During placement of concrete at the roof, the service load (lb) in a shore at Level 1 is most nearly:



- ☐ A. 4,930
 - ☐ B. 9,860
 - ☐ C. 11,650
 - ☐ D. 15,420
36. A contractor is planning to install a temporary structure that will support a concrete pump at a construction site. According to the construction contract, the contractor must submit design drawings and calculations for the temporary structure to the owner for approval. What is the purpose of this submittal requirement?
- ☐ A. To ensure compliance with building codes and regulations
 - ☐ B. To provide a record of the construction process
 - ☐ C. To demonstrate the contractor's experience and qualifications
 - ☐ D. To facilitate communication between the owner and contractor

PE CIVIL: STRUCTURAL PRACTICE EXAM

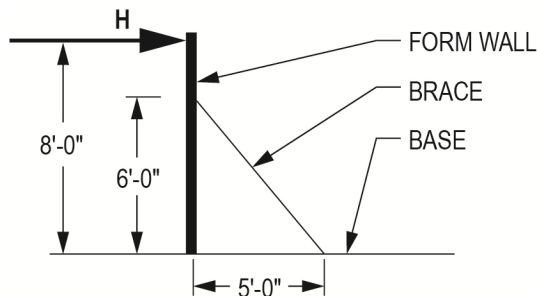
37. According to IBC 2018 and applicable quality assurance inspection requirements, which of the following inspection tasks are required for ASTM A325N high-strength bolts used in a steel-framed structure?

Select the **two** that apply.

- ☐ A. Verification of the magnitude of the clamping force
- ☐ B. Calibration of the torque wrench
- ☐ C. Confirmation that the proper fastener components are used
- ☐ D. Confirmation that the faying surfaces are brought into firm contact during installation of the bolts
- ☐ E. Confirmation that the bolt installer has an AWS D1.1 certification

38. To result in a compressive force of 15 kips on the diagonal brace shown, the horizontal force **H** (kips) on the top of the form wall shown must be _____.

Enter your response in the blank.



PE CIVIL: STRUCTURAL PRACTICE EXAM

39. A contractor is constructing a temporary stage with a maximum capacity of 500 people for a music festival. The stage is supported by steel wide-flange beams with a simple span of 20 ft. The beams are considered uniformly loaded. The safety requirements of the music festival state that all stage support beams have a maximum deflection of 1/4 in. or less. Assume the total uniform load is 780 lb/ft on each of the evenly spaced steel beam members.

Based on the uniform load, which of the following beam sizes meet the deflection requirements?

Select the **two** that apply.

- ☐ A. W10×49
- ☐ B. W12×26
- ☐ C. W12×53
- ☐ D. W14×22
- ☐ E. W14×43
- ☐ F. W16×26

40. A rough-terrain mobile crane is proposed for picking a 21-ton load on a jobsite. Refer to the load chart on the next page. The maximum radius (ft) at which this crane can safely pick this load is most nearly:

- ☐ A. 25
- ☐ B. 40
- ☐ C. 61
- ☐ D. 87

PE CIVIL: STRUCTURAL PRACTICE EXAM

40. (Continued)

Load chart

GRT655



Radius in Feet	Main Boom Length in Feet									
	34.9	43.7	52.5	61.4	70.2	79	87.8	96.7	105.5	114.3
8	110,000 (68.5)	—	—	—	—	—	—	—	—	—
10	88,750 (65)	81,000 (70.5)	79,700 (74)	79,200 (77)	72,650 (79.5)	—	—	—	—	—
12	80,150 (61)	78,100 (67.5)	76,400 (72)	75,050 (75)	68,200 (77.5)	*57,800 (80)	—	—	—	—
15	69,550 (55)	68,200 (63)	66,700 (68)	65,450 (72)	60,450 (75)	51,050 (77)	42,250 (79)	*34,500 (80)	—	—
20	55,300 (43)	55,800 (55.5)	54,900 (62)	53,900 (67)	49,400 (70.5)	42,050 (73)	34,950 (75.5)	33,300 (77)	28,100 (78.5)	*25,800 (80)
25	40,400 (26)	43,300 (46.5)	43,750 (55.5)	44,100 (61.5)	41,300 (66)	35,150 (69)	29,250 (72)	28,150 (74)	27,000 (76)	24,600 (77.5)
30	—	34,750 (35)	35,200 (48)	35,500 (56)	35,250 (61.5)	29,950 (65)	24,900 (68.5)	23,950 (71)	23,150 (73)	21,800 (75)
35	—	24,950 (17)	28,150 (39.5)	28,300 (50)	28,450 (56.5)	25,950 (61)	21,500 (64.5)	20,650 (67.5)	20,000 (70)	19,400 (72)
40	—	—	22,050 (29)	22,300 (43)	22,500 (51)	22,750 (56.5)	18,750 (61)	18,050 (64.5)	17,450 (67)	16,900 (69.5)
45	—	—	—	18,050 (34.5)	18,300 (45.5)	18,500 (52)	16,550 (57)	15,900 (61)	15,350 (64)	14,900 (66.5)
50	—	—	—	14,850 (24)	15,150 (38.5)	15,300 (47)	14,700 (53)	14,100 (57.5)	13,600 (61)	13,150 (64)
55	—	—	—	—	12,700 (30.5)	12,900 (41.5)	13,000 (48.5)	12,550 (53.5)	12,100 (57.5)	11,700 (61)
60	—	—	—	—	10,700 (19)	10,900 (35)	11,050 (43.5)	11,150 (49.5)	10,850 (54)	10,500 (58)
65	—	—	—	—	—	9300 (27)	9470 (38)	9560 (45)	9570 (50.5)	9420 (54.5)
70	—	—	—	—	—	7950 (14)	8100 (32)	8220 (40.5)	8240 (46.5)	8240 (51.5)
75	—	—	—	—	—	—	6950 (23.5)	7070 (35)	7120 (42.5)	7140 (48)
80	—	—	—	—	—	—	4650 (6.5)	6070 (29)	6150 (38)	6190 (44)
85	—	—	—	—	—	—	—	5210 (20.5)	5300 (32.5)	5370 (40)
90	—	—	—	—	—	—	—	—	4550 (26)	4640 (35.5)
95	—	—	—	—	—	—	—	—	3900 (17)	3990 (30)
100	—	—	—	—	—	—	—	—	—	3410 (23.5)
105	—	—	—	—	—	—	—	—	—	2890 (13.5)
Min. boom angle for indicated length (no load)										12°
Max. boom length at 0° boom angle (no load)										105.5 ft

NOTE: () Boom angles are in degrees.

*This capacity is based on maximum boom angle.

Boom Angle	Main Boom Length in Feet									
	34.9	43.7	52.5	61.4	70.2	79	87.8	96.7	105.5	114.3
0°	17,600 (27.1)	12,700 (35.9)	9480 (44.7)	7160 (53.6)	5420 (62.4)	4060 (71.2)	2970 (80)	2070 (88.9)	1320 (97.7)	—

NOTE: () Reference radii in feet.

80100630

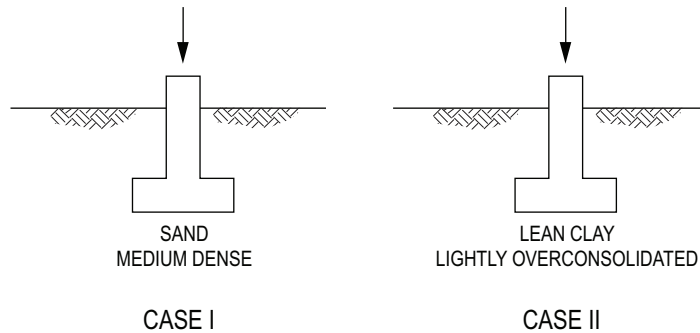
THIS CHART IS ONLY A GUIDE AND SHOULD NOT BE USED TO OPERATE THE CRANE.

The individual crane's load chart, operating instructions and other instructional plates must be read and understood prior to operating the crane

Grove GRT655/GRT655L | Page 9

PE CIVIL: STRUCTURAL PRACTICE EXAM

41. The figure shows two identical building footings with the same load but constructed in two different soil types. Which of the following statements is most correct?



- ☐ A. The long-term settlement for Case I is less than Case II.
 - ☐ B. The long-term settlement for Case II is less than Case I.
 - ☐ C. The long-term settlements are the same for both cases.
 - ☐ D. Settlement is not a concern for either case.
42. The most essential criteria for proper soil classification using the Unified Soil Classification System or the AASHTO Soil Classification System are:
- ☐ A. water content and soil density
 - ☐ B. Atterberg limits and specific gravity
 - ☐ C. grain-size distribution and water content
 - ☐ D. grain-size distribution and Atterberg limits

PE CIVIL: STRUCTURAL PRACTICE EXAM

43. The standard penetration test (SPT) is widely used as a simple and economic means of obtaining which of the following?
- ☐ A. A measurement of soil compressibility expressed in terms of a compression index
 - ☐ B. A direct measurement of the undrained shear strength
 - ☐ C. An indirect indication of the relative density of cohesionless soils
 - ☐ D. A direct measurement of the angle of internal friction
44. A department of transportation must remove and replace a 12-ft $\times$ 20-ft concrete slab on an interstate facility. To minimize disruption to traffic, the work must be completed during an 8-hr nighttime work shift. Nighttime temperatures average 50°F. If the minimum required compressive strength is 3,500 psi, the concrete mix most likely consists of:
- ☐ A. coarse aggregate, sand, Type II cement, chemical accelerator
 - ☐ B. sand, Type III cement, water, chemical accelerator
 - ☐ C. coarse aggregate, sand, Type V cement, water, chemical accelerator
 - ☐ D. coarse aggregate, sand, Type III cement, water, chemical accelerator
45. Fatigue in steel can be the result of:
- ☐ A. a reduction in strength due to cyclical loads
 - ☐ B. deformation under impact loads
 - ☐ C. deflection due to overload
 - ☐ D. expansion due to corrosion

PE CIVIL: STRUCTURAL PRACTICE EXAM

46. Sample concrete cylinders that are 6 inches in diameter and 12 inches high are tested to determine the compressive strength of the concrete f'_c . The test results are as follows:

Sample	Axial Compressive Failure Load (lb)
1	65,447
2	63,617
3	69,872

Based on the test results, the average 28-day compressive strength (psi) is most nearly:

- ☐ A. 615
- ☐ B. 2,250
- ☐ C. 2,340
- ☐ D. 2,470

PE CIVIL: STRUCTURAL PRACTICE EXAM

47. During testing of a sample in the laboratory, the following soil data are collected:

Combined weight of compacted soil sample and mold = 9.11 lb

Water content of soil sample = 11.5%

Weight of mold = 4.41 lb

Volume of mold = 0.03 ft³

The dry unit weight (pcf) of the soil sample is most nearly:

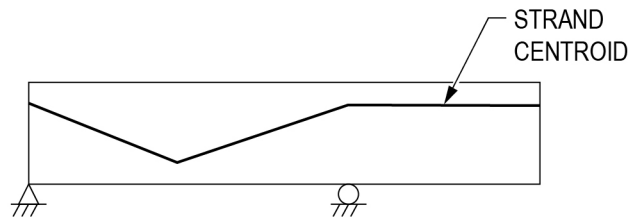
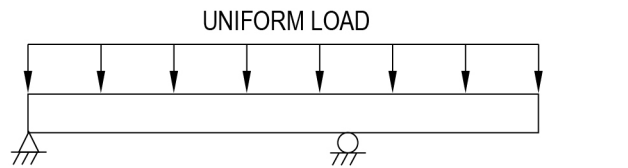
- ☐ A. 160
- ☐ B. 140
- ☐ C. 127
- ☐ D. 125

48. Normal-weight concrete with a maximum aggregate size of 1 1/2 in., $f'_c = 4,000$ psi, has moderate exposure to freezing and thawing. Neglecting tolerance, the recommended minimum air content is most nearly:

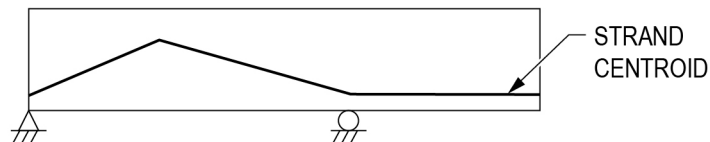
- ☐ A. 4.5%
- ☐ B. 5.0%
- ☐ C. 5.5%
- ☐ D. 6.0%

PE CIVIL: STRUCTURAL PRACTICE EXAM

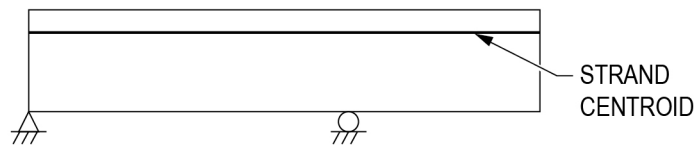
49. A loaded prestressed beam is shown in the figure. The most likely prestressing strand pattern would be:



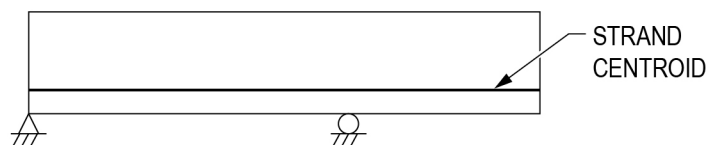
Option A



Option B



Option C



Option D

- ☐ A. Option A
- ☐ B. Option B
- ☐ C. Option C
- ☐ D. Option D

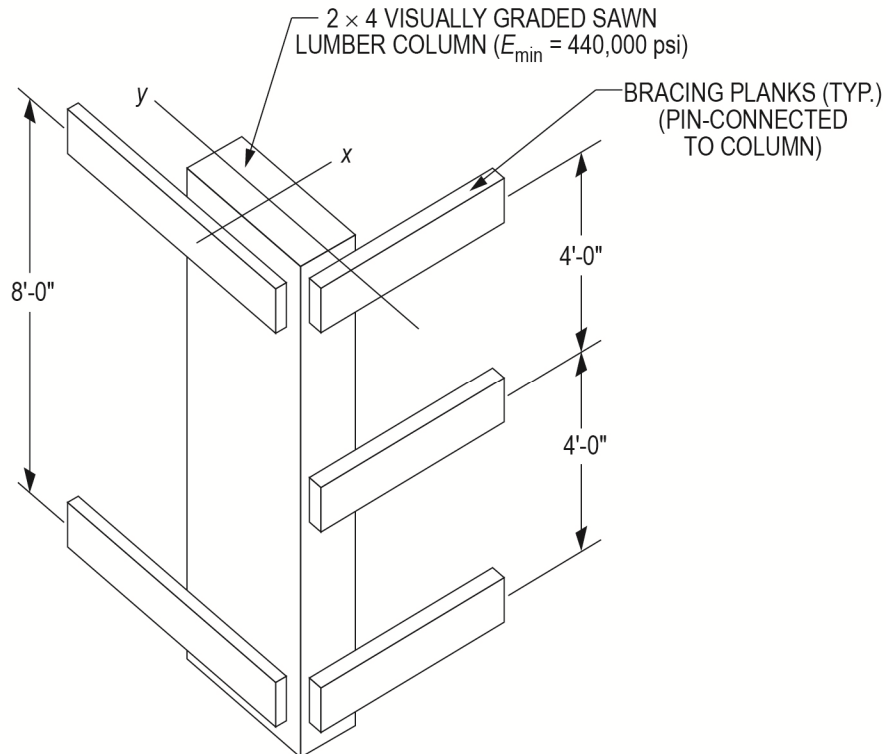
PE CIVIL: STRUCTURAL PRACTICE EXAM

50. The figure shows a timber column.

Design Code:

NDS: *National Design Specification for Wood Construction with Commentary*, 2018 edition
and *National Design Specification Supplement, Design Values for Wood Construction*, 2018 edition.

Assuming dry service, normal temperature conditions, and $C_i = 1.00$, the critical buckling design value F_{cE} (psi) using allowable stress design for the column is most nearly:



- ☐ A. 150
- ☐ B. 350
- ☐ C. 475
- ☐ D. 500

PE CIVIL: STRUCTURAL PRACTICE EXAM

51. A backfill material is described as sand-silt clay mix with plastic fines. The Unified Soil Classification is SM-SC. Per ASCE 7-16, the estimated design lateral soil load (psf/foot of depth) for relatively nonrigid walls is most nearly:

- ☐ A. 35
- ☐ B. 45
- ☐ C. 85
- ☐ D. 100

52. The first five 28-day average standard-cured concrete strength tests are shown below.

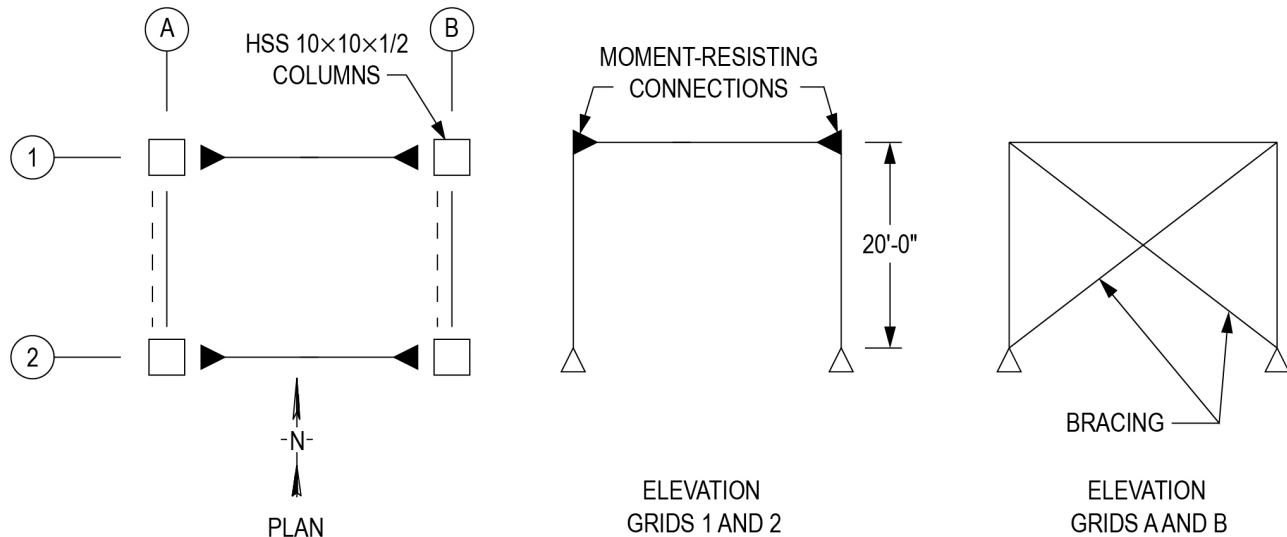
Test	28-Day Average f'_c	Date
1	4,215	March 3
2	4,160	March 7
3	3,625	March 11
4	4,010	March 12
5	4,015	March 16

The required $f'_c = 4,000$ psi. Which of the following statements is true?

- ☐ A. The concrete is unsatisfactory because the average of the last three consecutive tests is less than the required f'_c .
- ☐ B. The concrete is unsatisfactory because Test 3 did not meet the required f'_c .
- ☐ C. The concrete is satisfactory because the average of the first three tests is greater than or equal to the required f'_c .
- ☐ D. The concrete is satisfactory because Test 3 is within 500 psi of the required f'_c .

PE CIVIL: STRUCTURAL PRACTICE EXAM

53. A structure is shown in the figure. Considering the buckling limit state for Column A1, which of the following statements is true?



- A. Buckling in the north-south direction will control the column capacity.
- B. Buckling in the east-west direction will control the column capacity.
- C. Buckling in either direction might control the column capacity.
- D. Buckling will not control since both directions are stable.
54. Which of the following characterizes the typical failure of an overreinforced concrete beam that has a reinforcement ratio higher than the balanced ratio?
- A. The steel yields, and large deflections and tensile cracks will be observed prior to failure.
- B. The concrete crushes, and large deflections and tensile cracks will be observed prior to failure.
- C. The steel yields, and the beam fails suddenly without warning.
- D. The concrete crushes, and the beam fails suddenly without warning.

PE CIVIL: STRUCTURAL PRACTICE EXAM

55. The figure shows a wall construction plan.

Design Code:

NDS: *National Design Specification for Wood Construction with Commentary*, 2018 edition
and *National Design Specification Supplement, Design Values for Wood Construction*,
2018 edition.

Material:

Douglas Fir-Larch No. 2

Assumptions:

The gypsum board provides stud weak axis bracing.

$$C_D = 1.6$$

$$C_M = 1.0$$

$$C_t = 1.0$$

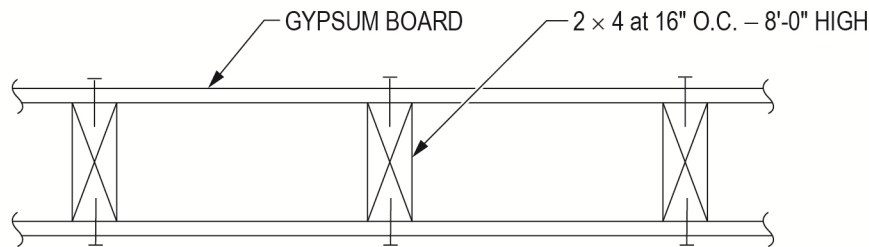
$$C_F = 1.15$$

$$C_i = 1.00$$

$$C_p = 0.33 \text{ strong axis}$$

$$C_p = 1.0 \text{ weak axis}$$

Using allowable stress design, the maximum vertical load (lb) per stud is most nearly:



- ☐ A. 2,700
- ☐ B. 3,300
- ☐ C. 3,700
- ☐ D. 4,300

PE CIVIL: STRUCTURAL PRACTICE EXAM

56. The following information applies to the masonry beam section shown:

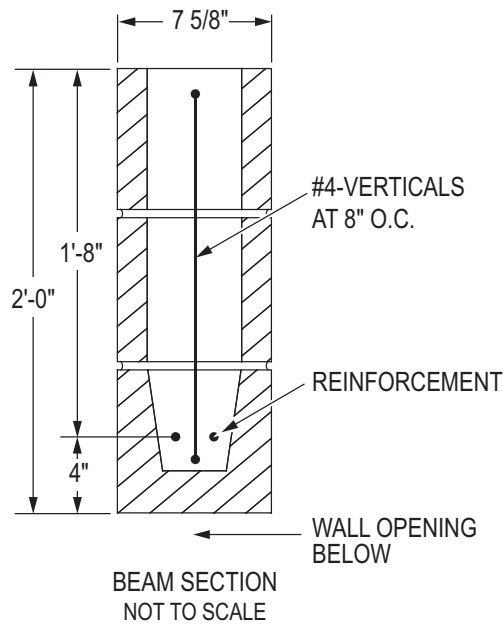
$$f'_m = 1,500 \text{ psi}$$

$$f_y = 60,000 \text{ psi}$$

Design Code:

TMS 402/602: *Building Code Requirements and Specification for Masonry Structures* (and related commentaries), 2016.

Use allowable stress design. Assume shear load is resisted by shear reinforcement only. The maximum allowable shear force (kips) for the beam is most nearly:



- ☐ A. 10
- ☐ B. 12
- ☐ C. 14
- ☐ D. 18

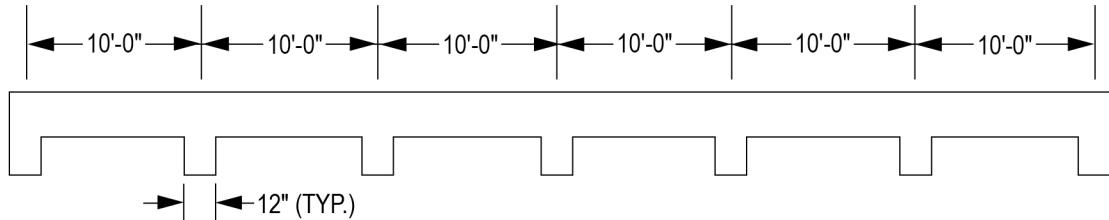
PE CIVIL: STRUCTURAL PRACTICE EXAM

57. A concrete slab of uniform thickness is continuous for six spans and is supported by 12-in.-wide concrete beams at 10 ft o.c. The following loads are applied to the slab:

DL = 100 psf (includes self-weight)

LL = 50 psf

Using ACI 318-14's alternate to frame analysis, the maximum factored negative moment in the slab (ft-kips/ft) is most nearly:



- ☐ A. 1.35
- ☐ B. 1.47
- ☐ C. 1.67
- ☐ D. 1.80

PE CIVIL: STRUCTURAL PRACTICE EXAM

58. The W10×22 steel beam ($F_y = 50$ ksi) shown in the figure is braced at midspan.

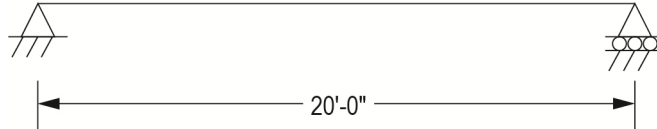
Work either the ASD or the LRFD version of the question. Assume $c_b = 1.0$.

ASD

The allowable flexural strength (ft-kips) of the beam is most nearly:

LRFD

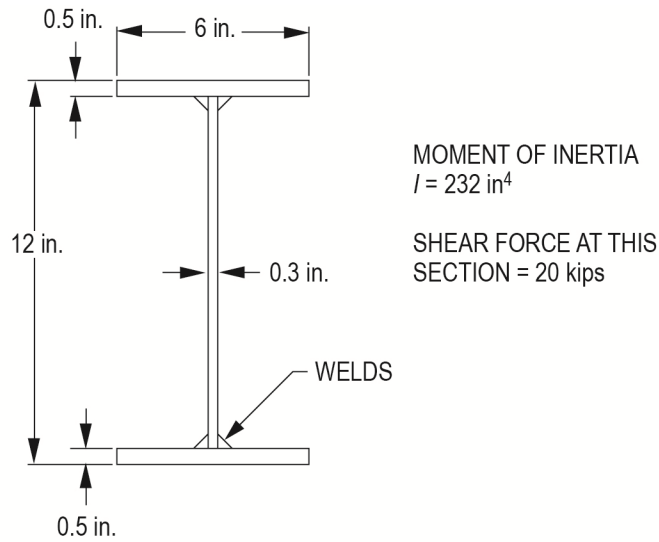
The design moment capacity ϕM_n (ft-kips) of the beam is most nearly:



- ☐ A. ASD = 45
LRFD = 68
- ☐ B. ASD = 51
LRFD = 76
- ☐ C. ASD = 56
LRFD = 84
- ☐ D. ASD = 65
LRFD = 97

PE CIVIL: STRUCTURAL PRACTICE EXAM

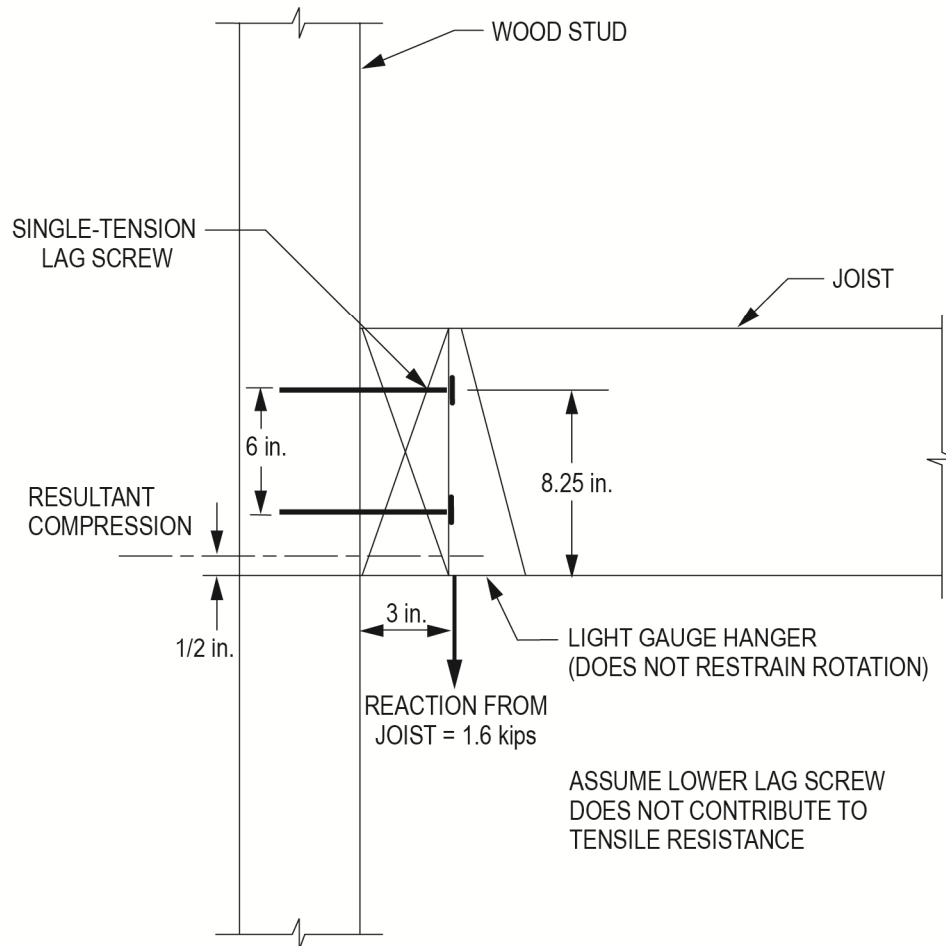
59. For the section shown, the unit shear force (kips/in.) in the welds connecting the top flange to the web (total unit shear in the two weld lines combined) is most nearly:



- ☐ A. 1.5
- ☐ B. 3.0
- ☐ C. 5.5
- ☐ D. 10.0

PE CIVIL: STRUCTURAL PRACTICE EXAM

60. A lag screw connection is shown in the figure. The tensile force (lb) in the top lag screw is most nearly:



- ☐ A. 582
- ☐ B. 619
- ☐ C. 800
- ☐ D. 2,400

PE CIVIL: STRUCTURAL PRACTICE EXAM

61. A connection is shown in the figure.

Design Code:

AISC: *Steel Construction Manual*, 15th edition, 2017.

Material:

A36 steel

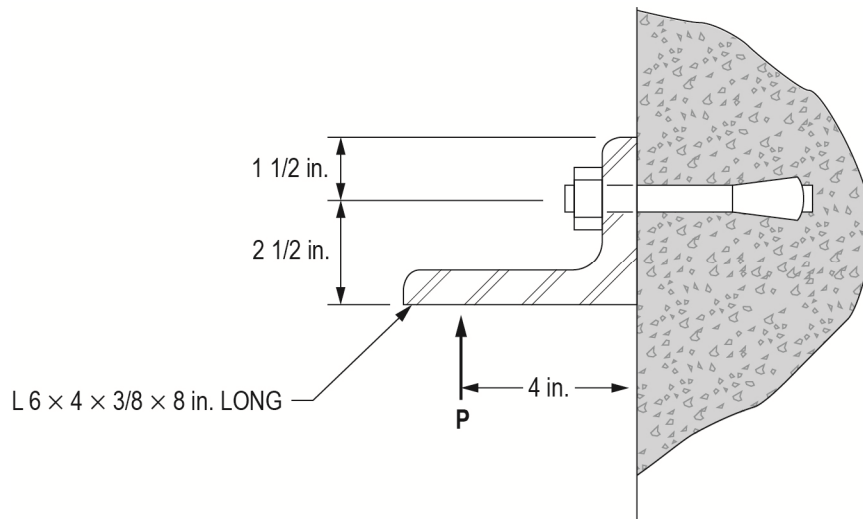
Assumptions:

The bolt is sufficient.

The concrete is sufficient.

The load P is equally distributed along the entire 8-in.-long angle.

The maximum load P (ASD) or ϕP_n (LRFD) in kips that can be carried by the angle is most nearly:



- ☐ A. ASD = 0.8
LRFD = 1.2
- ☐ B. ASD = 1.3
LRFD = 1.9
- ☐ C. ASD = 1.7
LRFD = 2.5
- ☐ D. ASD = 2.8
LRFD = 2.8

PE CIVIL: STRUCTURAL PRACTICE EXAM

62. Label the correct characteristics of each figure of a roof diagram.

CHARACTERISTICS

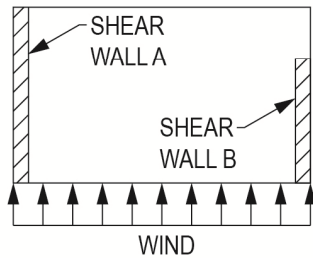
Flexible

Torsionally irregular

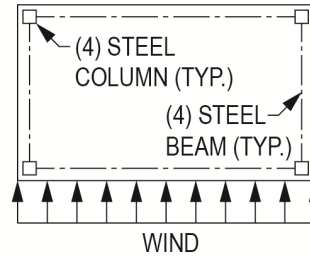
Rigid

Torsionally regular

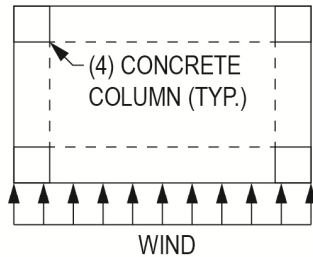
- WOOD
- DIAPHRAGM DEFLECTION CAN BE TOLERATED



- METAL DECK SUPPORTED BY STEEL BEAMS AND COLUMNS

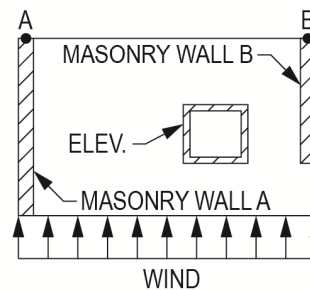


- PT CONCRETE SLAB SUPPORTED BY CONCRETE COLUMNS/BEAMS



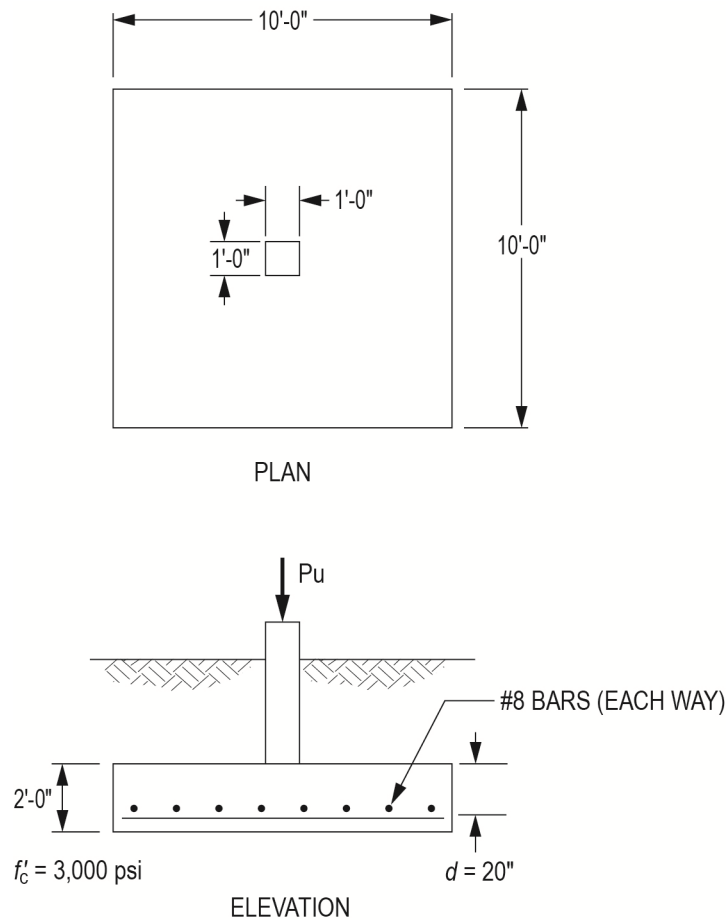
- CONCRETE SLAB SUPPORTED BY MASONRY WALLS

$$A_{\text{STORYDRIFT}} = 1.5 \left(\frac{A_{\text{STORYDRIFT}} + B_{\text{STORYDRIFT}}}{2} \right)$$



PE CIVIL: STRUCTURAL PRACTICE EXAM

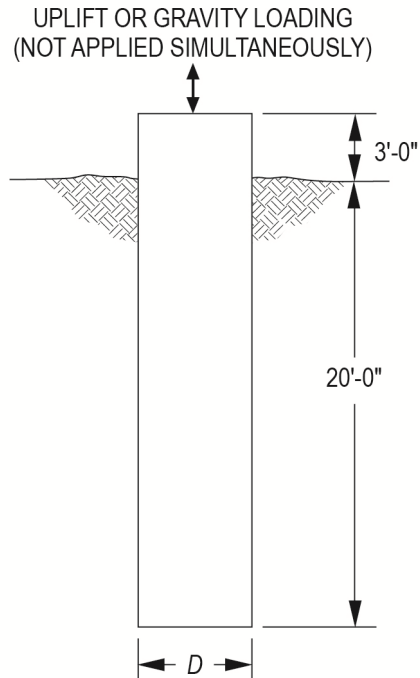
63. The figure shows a cast-in-place reinforced concrete spread footing for an interior column that is concentrically loaded. Punching shear controls the footing thickness in the design. Neglecting the shear strength of reinforcing, the design punching shear capacity of the footing (kips) per ACI 318-14, is most nearly:



- ☐ A. 870
- ☐ B. 680
- ☐ C. 560
- ☐ D. 420

PE CIVIL: STRUCTURAL PRACTICE EXAM

64. A concrete caisson is subjected to two separate loading conditions: a net uplift of 100 kips or a net gravity load of 75 kips. The loads are not applied simultaneously. The allowable soil skin friction is 200 psf, and the allowable soil bearing pressure is 4,000 psf. Neglecting the weight of the caisson, the required caisson diameter (ft) is most nearly:



- ☐ A. 8
- ☐ B. 7
- ☐ C. 6
- ☐ D. 5

PE CIVIL: STRUCTURAL PRACTICE EXAM

65. Assuming a bearing-type connection, the load capacity (kips) for a 1-in.-diameter A307 through bolt in double shear is most nearly:

- A. ASD = 10.6
LRFD = 15.9
- B. ASD = 21.2
LRFD = 31.9
- C. ASD = 28.4
LRFD = 53.0
- D. ASD = 56.9
LRFD = 56.7

66. The following information applies to the structure shown in the figure:

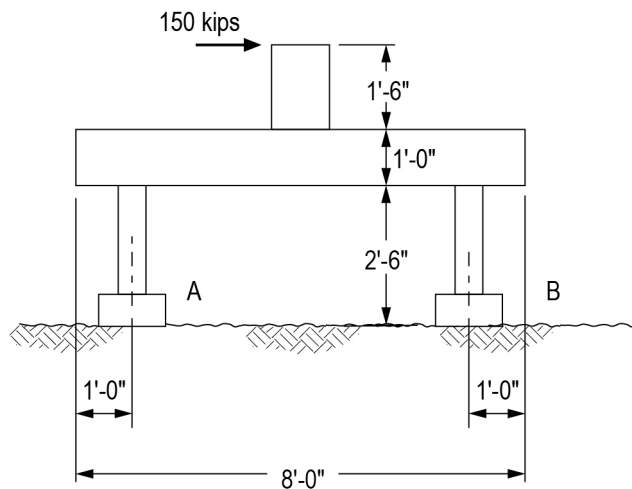
Neglect beam weight.

Beam is infinitely stiff.

Footing size is 24 in. $\times$ 24 in. at each footing.

Soil vertical modulus of subgrade, $K = 100 \text{ lb/in}^3$

The vertical settlement (in.) of the footing at B is most nearly:



- A. 2.2
- B. 1.6
- C. 1.1
- D. 0.7

PE CIVIL: STRUCTURAL PRACTICE EXAM

67. The cross section for a 50-ft-span, rectangular, prestressed beam is shown in the figure. The beam has no mild reinforcing steel.

Design Code:

PCI: *PCI Design Handbook: Precast and Prestressed Concrete*, 7th edition, 2010.

Design data for prestressing strands:

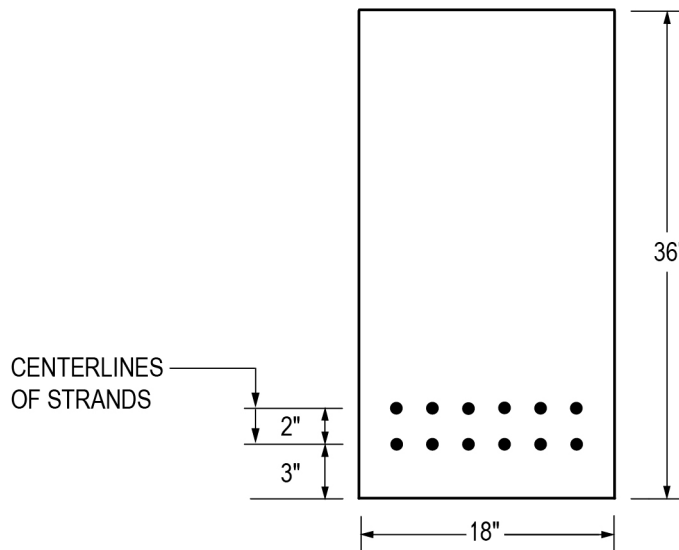
Low relaxation – 1/2-in. diameter

$$f_{pu} = 270 \text{ ksi}$$

$$A_s = 0.153 \text{ in}^2 \text{ per strand}$$

Stress at release = 175 ksi per strand (after initial losses)

If the top fiber stress at the midspan of the beam due to the beam self-weight is 0.65 ksi, the total top fiber stress (ksi) at release at midspan is most nearly:



- ☐ A. 0.01 (tension)
- ☐ B. 0.50 (tension)
- ☐ C. 0.99 (compression)
- ☐ D. 2.29 (compression)

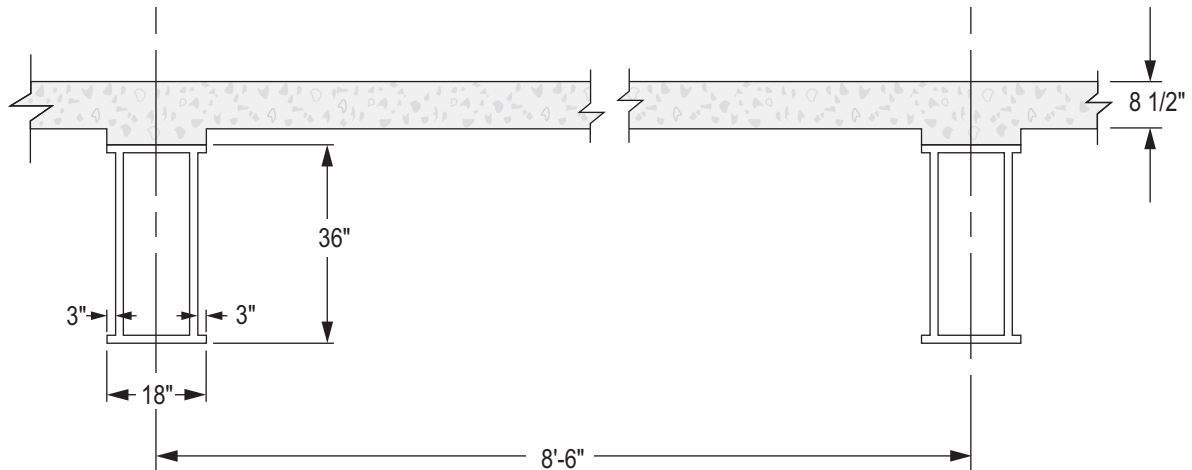
PE CIVIL: STRUCTURAL PRACTICE EXAM

68. The figure shows a cross section of a deck slab of a steel girder bridge.

Design Code:

AASHTO LRFD Bridge Design Specifications, 8th edition, 2017.

The effective span length L_{eff} to be used for the empirical design of the deck slab is most nearly:



- ☐ A. 6'-9"
- ☐ B. 7'-0"
- ☐ C. 7'-6"
- ☐ D. 8'-6"

69. A footing with the plan dimensions of 8 ft $\times$ 8 ft is being cast on a sandy gravel and gravel (GW and GP) material. No lateral loads and no vertical load eccentricity are present. Based on the IBC 2018 presumptive load bearing value, the allowable vertical load (kips) that can be placed on the foundation is _____.

Enter your response in the blank.

PE CIVIL: STRUCTURAL PRACTICE EXAM

70. According to IBC 2018, which of the following represent the **minimum** factor of safety against uplift of deep pile elements due to wind or seismic loading?

Select the **two** that apply.

- ☐ A. One and one-half when capacity is determined by approved load tests
- ☐ B. Two when capacity is determined by approved load tests
- ☐ C. Two when capacity is determined by an approved method of analysis
- ☐ D. Three when capacity is determined by an approved method of analysis
- ☐ E. Four when capacity is determined by an approved method of analysis

71. According to IBC 2018, which of the following represent the **minimum** compressive strength of a precast concrete pile before driving?

Select the **two** that apply.

- ☐ A. 75% of the specified compressive strength (f'_c)
- ☐ B. 80% of the specified compressive strength (f'_c)
- ☐ C. Not less than the strength sufficient to withstand handling and driving forces
- ☐ D. One and one-half times the sufficient strength to withstand initial prestress load upon transfer
- ☐ E. Must be determined by load testing

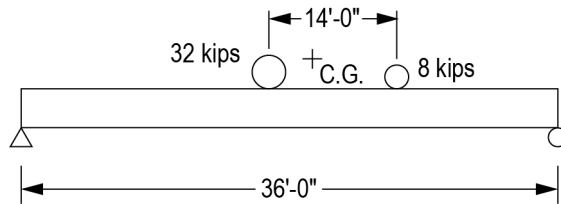
72. According to AISC *Steel Construction Manual*, 15th edition, the resistance factor $\phi = 0.90$ and the safety factor $\Omega = 1.67$ apply to which of the following limit states?

Select the **four** that apply.

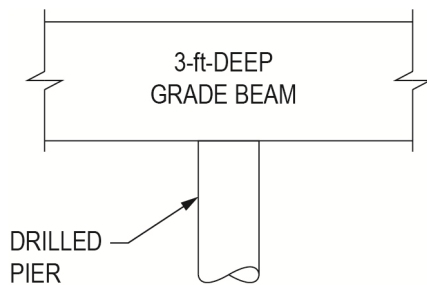
- ☐ A. Tensile yielding in the gross section
- ☐ B. Tensile rupture in the net section
- ☐ C. Compressive failure including consideration of all buckling modes
- ☐ D. Flexural failure including consideration of all buckling modes
- ☐ E. Shear failure including consideration of all buckling modes

PE CIVIL: STRUCTURAL PRACTICE EXAM

73. The beam shown is loaded by a set of moving loads. The center of gravity (C.G.) of the moving load is 2.8 ft from the 32-kip load. The maximum moment in the beam (ft-kips) is _____.



74. For the figure shown, factored vertical reactions from the grade beam to the drilled pier are $LL = 10$ kips and $DL = 8$ kips. According to ASCE 7-16, the **minimum** horizontal design strength (lb) of the connection from the grade beam to the drilled pier is most nearly:



- ☐ A. 1,800
- ☐ B. 1,280
- ☐ C. 900
- ☐ D. 500

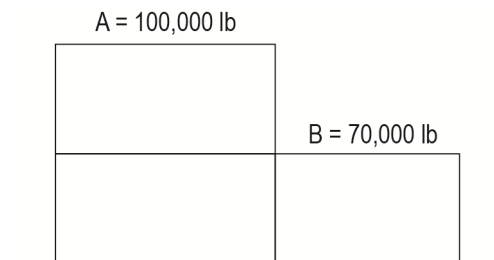
PE CIVIL: STRUCTURAL PRACTICE EXAM

75. For a painted 3/4-in. plate bolted to a CMU wall with two 5/8-in.-diameter bolts in short-slotted holes, which of the following are true?

Select the **three** that apply.

- ☐ A. Minimum edge distance, center of hole to edge of plate = 1 in.
- ☐ B. Minimum edge distance, center of hole to edge of plate = 7/8 in.
- ☐ C. Minimum center-to-center spacing of holes = 1.875 in.
- ☐ D. Minimum center-to-center spacing of holes = 1.670 in.
- ☐ E. Maximum edge distance, center of hole to edge of plate = 9 in.
- ☐ F. Maximum edge distance, center of hole to edge of plate = 6 in.

76. The weight of a structure is as shown. $S_{DS} = 0.439$ g. Per ASCE 7-16, the design strength (lb) of the connection transmitting seismic forces from the low roof to the main structure is most nearly:



- ☐ A. 3,500
- ☐ B. 4,100
- ☐ C. 8,500
- ☐ D. 9,900

PE CIVIL: STRUCTURAL PRACTICE EXAM

77. An expansion anchor with medium reliability is governed by concrete pullout experiencing tension loads and will be anchored in a plain concrete footing with no supplemental reinforcement. The strength reduction factor ϕ is most nearly:

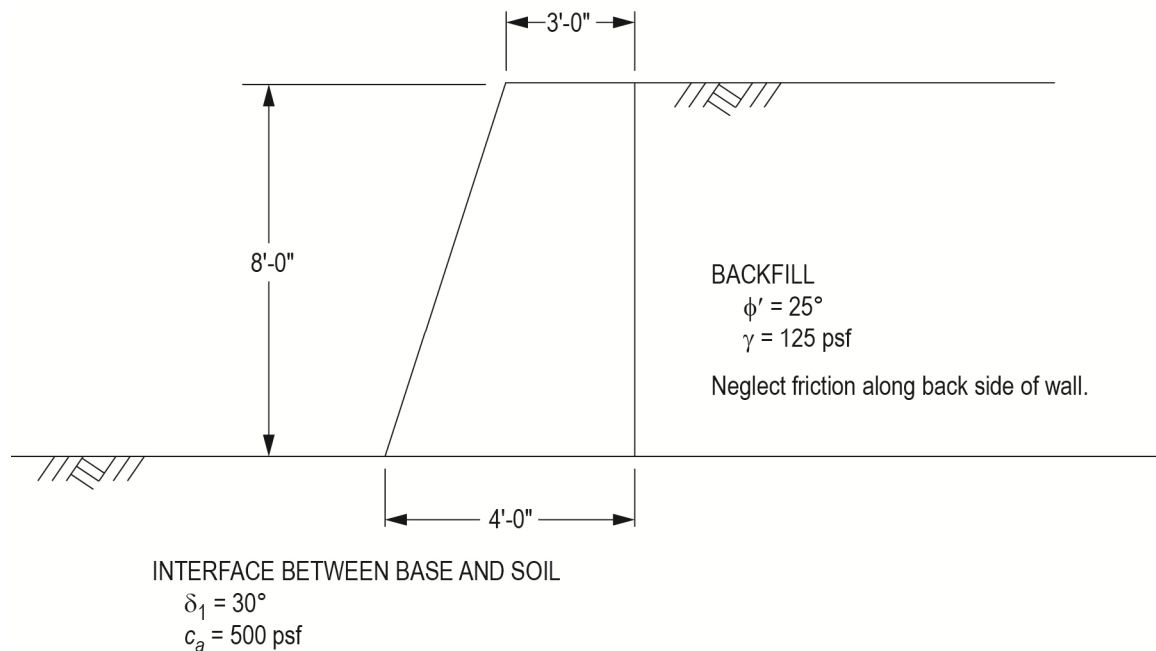
☐ A. 0.75
☐ B. 0.7
☐ C. 0.65
☐ D. 0.55

78. The nominal strength (lb) of a #6 Grade 60 rebar anchor in tension must not exceed _____.

Enter your response in the blank.

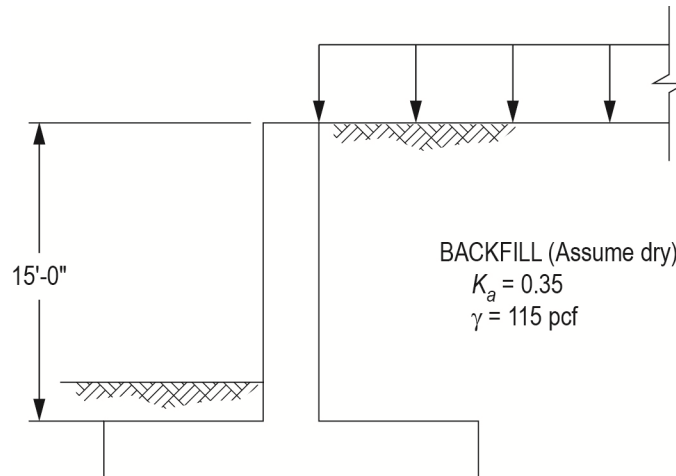
79. For the reinforced concrete gravity retaining wall shown, the factor of safety against sliding is _____. Assume $\delta_{RC} = 150$ pcf.

Enter your response in the blank.



PE CIVIL: STRUCTURAL PRACTICE EXAM

80. For the reinforced concrete retaining wall shown, the total active earth pressure force (plf) per unit width of wall is most nearly:



- ☐ A. 5,840
- ☐ B. 8,280
- ☐ C. 10,400
- ☐ D. 12,800

PE CIVIL: STRUCTURAL SOLUTIONS

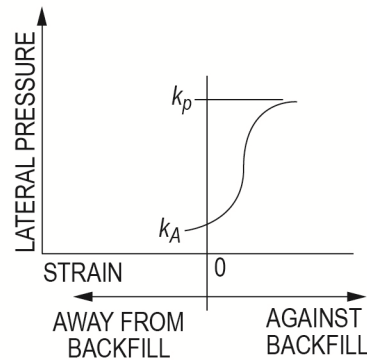
PE CIVIL: STRUCTURAL SOLUTIONS

1. $1,000 \text{ kN} = 1,000 \text{ kN} \times \frac{1 \text{ ton}}{8.896444 \text{ kN}} = 112.4 \text{ tons}$

$150 \text{ tons} > 112.4 \text{ tons}$

THE CORRECT ANSWER IS: C

2. The wall translation (or strain) required to achieve the passive state is at least twice that required to reach the active state.



THE CORRECT ANSWER IS: D

3. Effective vertical stress at Point A, σ'_v
 $= 10 \text{ ft} \times 120 \text{ pcf} + 5 \text{ ft} (120 \text{ pcf} - 62.4 \text{ pcf}) + 7 \text{ ft} (110 \text{ pcf} - 62.4 \text{ pcf})$
 $= 1,200 \text{ psf} + 288 \text{ psf} + 333 \text{ psf}$
 $= 1,821 \text{ psf}$

THE CORRECT ANSWER IS: B

4. The ultimate bearing capacity would be based on buoyant unit weight, also referred to as the effective unit weight.

Effective unit weight = saturated unit weight – unit weight of water

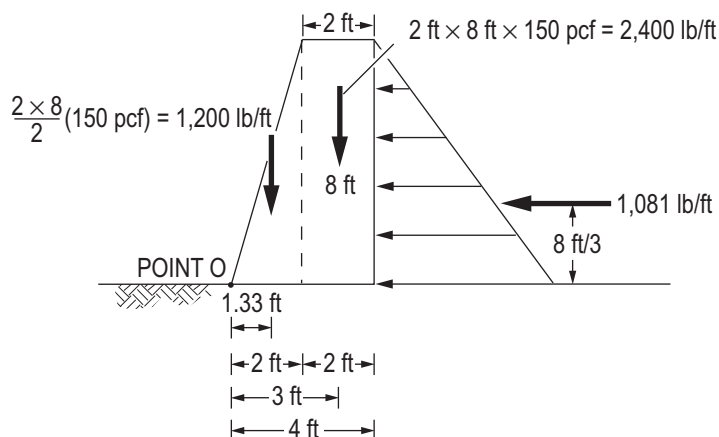
THE CORRECT ANSWER IS: A

5. Because the structure is cantilevered, in addition to the wind, dead load and live load will contribute to uplift.

THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

6. $\phi = 32^\circ$ $K_a = \tan^2(45 - \phi/2) = 0.307$
 $\gamma_t = 110 \text{ pcf}$ $P_a = (0.5)(110)(8)^2(0.307) = 1,081 \text{ lb/ft}$
 $M_a = (1,081)(8/3) = 2,883 \text{ ft-lb/ft}$
 $(2)(8)(150)(1)(3) = 7,200 \text{ ft-lb/ft}$
 $(1/2)(2)(8)(150)(1)(2)(2/3) = 1,600 \text{ ft-lb/ft}$ } total = 8,800 ft-lb/ft
 $SF = 8,800/2,883 = 3.05$

**THE CORRECT ANSWER IS: A**

7. Reference: ASCE 7-16

$$p_f = 0.7 C_e C_t I_s p_g$$

Eq. 7.3-1

where $C_e = 0.9$

Table 7.3-1

$$C_t = 1.2$$

Table 7.3-2

$$I_s = 0.8$$

Table 1.5-2

$$p_g = 20 \text{ psf}$$

$$p_f = (0.7)(0.9)(1.2)(0.8)(20)$$

$$= 12.1 \text{ psf}$$

$$p_s = C_s p_f \text{ (Eq. 7.4-1) and } C_s = 1.0 \text{ (Fig. 7.4-1)}$$

$$= 1.0(12.1) = 12.1$$

$$\text{Unbalanced snow load} = I_s p_g$$

Par. 7.6.1

$$= 0.8(20)$$

$$= 16 \text{ psf}$$

16 psf controls

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

8. $\frac{60(200)}{2} = 6 \text{ kips}$

THE CORRECT ANSWER IS: B

9. Reference: IBC 2018/ASCE 7-16. (IBC 2018 refers to ASCE 7-16.)
Per ASCE 7-16, Section 11.7 (Seismic Design Category A)

$$\begin{aligned} F &= 0.01w & w &= \text{dead load} & (\text{Eq. 1.4-1}) \\ &= 0.01(200,000) \\ &= 2,000 \text{ lb} \end{aligned}$$

THE CORRECT ANSWER IS: A

10. Maximum moment occurs when load is at beam midspan.

$$M = \frac{PL}{4} = \frac{(20 \text{ kips})(30 \text{ ft})}{4} = 150 \text{ ft-kips}$$

THE CORRECT ANSWER IS: 149 to 151

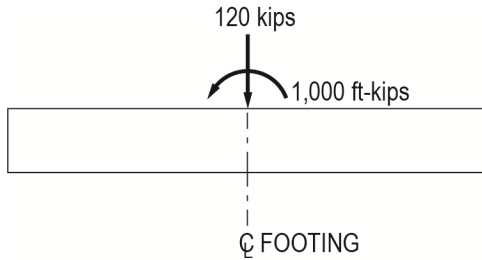
11. Take moments about Support B:

$$\begin{aligned} R_a \times 24 - 6.4 \times 24 - 2 \times 24^2/2 + 20 + 1.55 \times 8^2/2 + 1 \times 8 &= 0 \\ R_a &= 28 \text{ kips} \end{aligned}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

12. Weight of footing = $2.5 \text{ kips/ft} (24 \text{ ft}) = 60 \text{ kips}$
 $+ \uparrow \sum F_v = -80 \text{ kips} - 60 \text{ kips} + 20 \text{ kips} = -120 \text{ kips}$
 $+ \curvearrowright \sum M_{\text{CL}} = -80 \text{ kips} (10 \text{ ft}) - 20 \text{ kips} (10 \text{ ft}) = -1,000 \text{ ft-kips}$



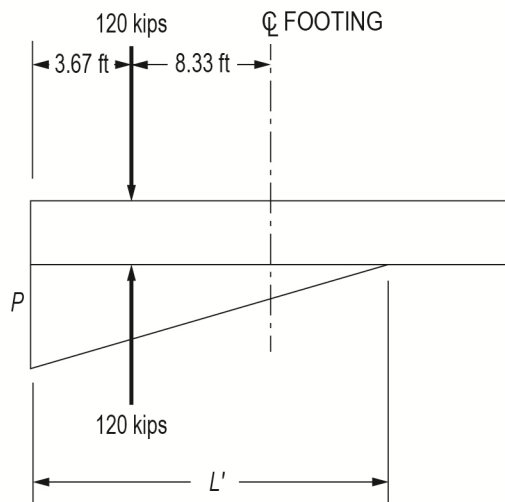
$$e = \frac{1,000 \text{ ft-kips}}{120 \text{ kips}}$$

$$e = 8.33 \text{ ft from CL}$$

$$\frac{L}{6} = \frac{24}{6} = 4 \text{ ft} < 8.33 \text{ ft}$$

Resultant not within kern

$\therefore$ Footing not fully effective



Resultant of soil pressure must align with eccentric resultant load.

L' is the effective length of the triangular soil pressure pattern.

$$\therefore 3.67 \text{ ft} = \frac{L'}{3} \rightarrow L' = 3(3.67 \text{ ft}) = 11.01 \text{ ft}$$

$$\frac{1}{2} PL' = \frac{1}{2} P(11.01 \text{ ft}) = 120 \text{ kips} \rightarrow P = \frac{2(120 \text{ kips})}{11.01 \text{ ft}} = 21.8 \text{ kips/ft}$$

$$P_{\text{max}} = \frac{P}{8 \text{ ft}} = \frac{21.8 \text{ kips/ft}}{8 \text{ ft}} = 2.72 \text{ kips/ft}^2$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

13. Reference: IBC 2018

Design spectral response acceleration at short periods: 1613.2.4

$$S_{DS} = \frac{2}{3} S_{MS} \quad (\text{Equation 16-38})$$

Adjusted maximum considered earthquake spectral response acceleration for short periods: 1613.2.3

$$S_{MS} = F_a S_s \quad (\text{Equation 16-36})$$

For Site Class D and mapped short period maximum considered earthquake spectral acceleration: Table 1613.2.3(1)

$$S_s = 100\% g$$

$$F_a = 1.1$$

$$\therefore S_{MS} = 1.1 \times 1.0 = 1.1$$

$$S_{DS} = \frac{2}{3} \times 1.1 = 0.73$$

THE CORRECT ANSWER IS: B

14. Reference: ASCE 7-16, p. 268

26.10.2 Velocity Pressure. Velocity pressure, q_z , evaluated at height z above ground shall be calculated by the following equation:

$$q_z = 0.00256 K_z K_{zt} K_d K_e V^2 \text{ (lb/ft}^2\text{); } V \text{ in mph} \quad (26.10-1)$$

$$q_z = 0.00256 K_z K_{zt} K_d K_e V^2 = 0.00256 (0.9) (1.0) (0.85) (1.0) (120)^2 = 28.2 \text{ psf}$$

Note: $K_h = K_z$ per 26.10.1

THE CORRECT ANSWER IS: 28.0 to 28.4

15. $w = (20 \text{ lb/ft}^2)(8 \text{ ft}) = 160 \text{ lb/vertical ft per brace location}$

$$\sum M_a = 0$$

$$\sum M_a = (160 \text{ lb/ft})(16 \text{ ft})(16 \text{ ft}/2) - 10 \text{ ft} (R_x) = 0$$

$$R_x = 2,048 \text{ lb}$$

$$\text{Axial load in brace} = \frac{(2,048)\sqrt{2}}{1} = 2,896 \text{ lb}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

16. By inspection, Member b = 0 kips, and Member c = 100 kips.

THE CORRECT ANSWER IS: B

17. Beam stress, $f = M/S$, where $M = wL^2/8$ and $S = bh^2/6$.

S is equal for both beams, but M varies because it depends on beam length.

Beam 1 (shorter beam): $M_1 = wL^2/8$

Beam 2 (longer beam): $M_2 = w(2L)^2/8 = 4wL^2/8$

M_2 is four times greater than M_1 . Therefore, the maximum bending stress is four times greater in the longer beam.

THE CORRECT ANSWER IS: D

18. Uniform load: $V = \frac{wL}{2} = \frac{1(30)}{2} = \frac{30 \text{ kips}}{2} = 15 \text{ kips}$

Point load: $V = \frac{P}{2} = 15 \text{ kips}$

$P = 2(15) = 30 \text{ kips}$

THE CORRECT ANSWER IS: C

19. I_x is maximum for this section by inspection, or calculate $I_x \approx \sum Ad^2$ for each section.

THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

20. Calculate reactions by taking moments about Support A:

$$(6 \text{ kips})(5 \text{ ft}) + (1 \text{ kip/ft})(20 \text{ ft})(20 \text{ ft}/2) = R_B(20 \text{ ft})$$

$$R_B = 11.5 \text{ kips}; R_A = 14.5 \text{ kips}$$

Location of maximum moment is where shear diagram = 0

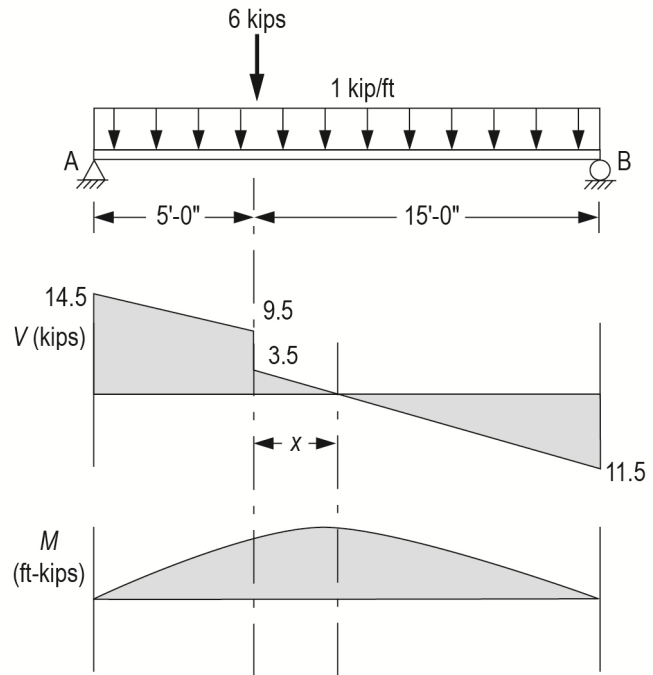
$$\frac{3.5 \text{ kips}}{x} = \frac{11.5 \text{ kips}}{(15 \text{ ft} - x)}$$

$$3.5(15 - x) = 11.5x$$

$$x = 3.5 \text{ ft}$$

Distance from right end:

$$20 \text{ ft} - (5 \text{ ft} + 3.5 \text{ ft}) = 11.5 \text{ ft}$$



THE CORRECT ANSWER IS: D

21.
$$FEM_{BA} = \frac{Pa^2b}{L^2} = \frac{(2)(12)^2(8)}{20^2} = 5.76 \text{ ft-kips}$$

$$FEM_{BC} = \frac{wL^2}{12} = \frac{(1)(15)^2}{12} = 18.75 \text{ ft-kips}$$

$$18.75 - 5.76 = 12.99 \text{ ft-kips}$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

22.
$$n = \frac{E_{\text{steel}}}{E_{\text{wood}}} = \frac{29 \times 10^6}{1.6 \times 10^6} = 18.125$$

$$A'_{\text{steel}} = nwh = (18.125)(6)(0.125) = 13.6 \text{ in}^2$$

$$A_{\text{wood}} = wh = (6)(8) = 48 \text{ in}^2$$

$$A = A'_{\text{steel}} + A_{\text{wood}} = 13.6 + 48 = 61.6 \text{ in}^2$$

$$\sum \bar{y}A = (\bar{y}_{\text{steel}} \times A'_{\text{steel}}) + (\bar{y}_{\text{wood}} \times A_{\text{wood}}) = (0.0625)(13.6) + (4.125)(48) = 198.85 \text{ in}^3$$

$$c = \bar{y} = \frac{\sum \bar{y}A}{A} = \frac{198.85}{61.6} = 3.23 \text{ in.}$$

$$I_{\text{wood}} = \frac{b \times d^3}{12} + [A_{\text{wood}} \times (\bar{y}_{\text{wood}} - \bar{y})^2] = \frac{(6)(8)^3}{12} + (48)(4.125 - 3.23)^2 = 294.4 \text{ in}^4$$

$$I_{\text{steel}} = \frac{b' \times d^3}{12} + [A'_{\text{steel}} \times (\bar{y} - \bar{y}_{\text{steel}})^2]$$

$$= \frac{(18.125)(6)(0.125)^3}{12} + (13.6)(3.23 - 0.0625)^2 = 136.5 \text{ in}^4$$

$$I = I_{\text{wood}} + I_{\text{steel}} = 294.4 + 136.5 = 430.9 \text{ in}^4$$

$$S_{\text{bottom}} = \frac{I}{c} = \frac{430.9}{3.23} = 133 \text{ in}^3$$

$$M = \frac{P \times a \times b}{L} = \frac{(6)(4)(6)}{10} = 14.4 \text{ ft-kips}$$

$$f'_b = \frac{M}{S} = \frac{(14.4)(12 \text{ in./ft})}{133} = 1.3 \text{ ksi}$$

$$f_b = f'_b \times n = (1.3)(18.125) = 23.5 \text{ ksi}$$

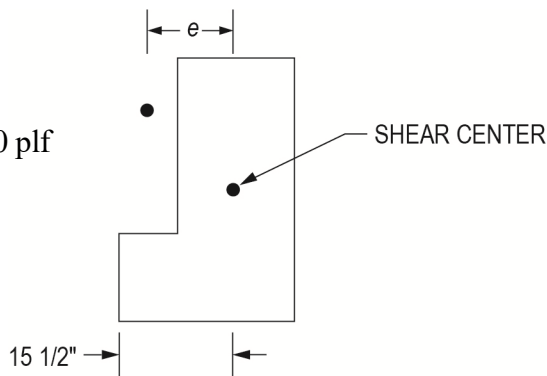
THE CORRECT ANSWER IS: D

23. Reference: ACI 318-14
 $15.5 \text{ in.} - 6 \text{ in.} = 9.5 \text{ in.}$

$$W_{u/DT} = [1.2(10 \text{ psf}) + 1.6(50 \text{ psf})]15 \text{ ft} = 1,380 \text{ plf}$$

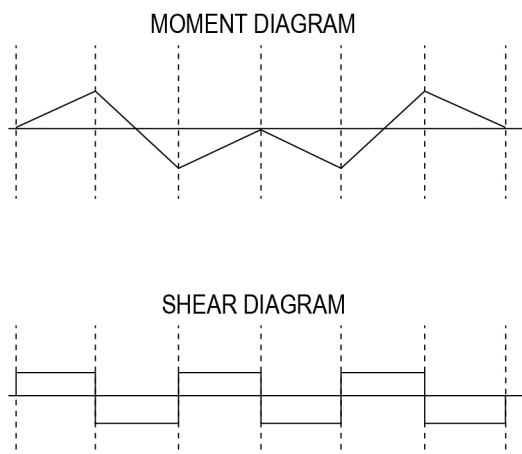
$$R_{u/DT} = \frac{1}{2}(1.38 \text{ klf})(60 \text{ ft}) = 41.4 \text{ kips}$$

$$T_u = 41.4 \text{ kips} \left(\frac{9.5 \text{ in.}}{12 \text{ in./ft}} \right) = 32.8 \text{ ft-kips}$$

**THE CORRECT ANSWER IS: D**

PE CIVIL: STRUCTURAL SOLUTIONS

24. Referring to the beam shown in the question, there is a zero moment at the hinge.



THE CORRECT DIAGRAMS ARE SHOWN ABOVE.

25.
$$M_{\max} = \frac{(200 \text{ lb/ft})(20 \text{ ft})^2}{8} + \frac{(1,000 \text{ lb})(20 \text{ ft})}{4}$$

$$= 10,000 \text{ ft-lb} + 5,000 \text{ ft-lb} = 15,000 \text{ ft-lb}$$

$$I = \frac{(4 \text{ in.})(12 \text{ in.})^3}{12} = 576 \text{ in}^4$$

$$f_b = \frac{Mc}{I}$$

$$= \frac{(15,000 \text{ ft-lb})(6 \text{ in.})}{576 \text{ in}^4} (12 \text{ in./ft})$$

$$f_b = 1,875 \text{ psi}$$

THE CORRECT ANSWER IS: C

26.
$$I = \frac{bd^3}{12} - \frac{b_1d_1^3}{12} = \frac{8(12)^3 - 4(8)^3}{12} = 981 \text{ in}^4$$

$$V = 1,000 \text{ plf} \left(\frac{10 \text{ ft}}{2} \right) + \left(\frac{4,000 \text{ lb}}{2} \right) = 7,000 \text{ lb}$$

$$v_{\text{horizontal}} = \frac{VQ}{IB} = \frac{7,000 \text{ lb} (85 \text{ in}^3)}{981 \text{ in}^4 (2 \times 2 \text{ in.})} = 151 \text{ psi}$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

27. $M_S = (20 \text{ kips})(12 \text{ in.} - 5 \text{ in.}) = 140 \text{ in.-kips}$

$N_S = 20 \text{ kips}$

$$\sigma_{\text{flexure}} = \frac{(140 \text{ in.-kips})(5 \text{ in.})}{90 \text{ in}^4} = 7.8 \text{ ksi} \begin{cases} \text{Top tension} \\ \text{Bottom compression} \end{cases}$$

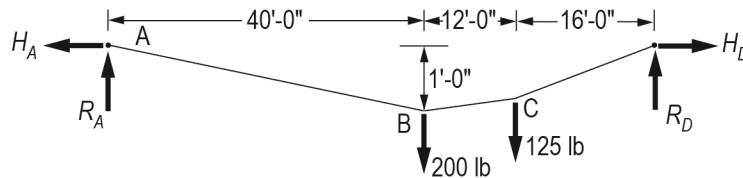
$$\sigma_{\text{axial}} = \frac{20 \text{ kips}}{7 \text{ in}^2} = 2.9 \text{ ksi tension}$$

Max compressive stress (at bottom):

$$-7.8 \text{ ksi} + 2.9 \text{ ksi} = -4.9 \text{ ksi}$$

THE CORRECT ANSWER IS: B

28.



$$R_A = \frac{200 \text{ lb} \times 28 \text{ ft} + 125 \text{ lb} \times 16 \text{ ft}}{68 \text{ ft}} = 111.76 \text{ lb}$$

$$R_D = \frac{200 \text{ lb} \times 40 \text{ ft} + 125 \text{ lb} \times 52 \text{ ft}}{68 \text{ ft}} = 213.24 \text{ lb}$$

$$H_A = \frac{40 \text{ ft}}{1 \text{ ft}} \times R_A = 4,470.4 \text{ lb}$$

$$H_A = H_D$$

$$F_{CD} = \sqrt{213.24^2 + 4,470.4^2}$$

$$F_{CD} = 4,475.5 \text{ lb}$$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

29. Reference: AISC, 15th ed., Beam Design

$$\Delta_{\text{conc. load}} = \frac{23 PL^3}{648 EI}$$

$$\Delta_{\text{uniform}} = \frac{5 WL^4}{384 EI} \Rightarrow \text{Set } \Delta_{\text{conc. load}} + \Delta_{\text{uniform}} = L/240$$

$$I \left(\frac{L}{240} \right) = \frac{23 PL^3}{648 E} + \frac{5 WL^4}{384 E}$$

$$I_{\text{required}} = \frac{240}{L} \left[\frac{23 PL^3}{648 E} + \frac{5 WL^4}{384 E} \right]$$

$$= \frac{240}{(15)(12)} \left[\frac{(23)30(15 \times 12)^3}{648 (29,000)} + \frac{5 (1/12)(15 \times 12)^4}{384 (29,000)} \right]$$

$$= 1.33(214.1 + 39.3) = 337 \text{ in}^4$$

- A. W12×40 $I = 307 \text{ in}^4$
- B. W12×45 $I = 348 \text{ in}^4 > 337$ OK
- C. W12×50 $I = 391 \text{ in}^4 > 337$ OK
- D. W14×26 $I = 245 \text{ in}^4$
- E. W14×30 $I = 291 \text{ in}^4$
- F. W14×34 $I = 340 \text{ in}^4 > 337$ OK
- G. W16×26 $I = 301 \text{ in}^4$
- H. W16×31 $I = 375 \text{ in}^4 > 337$ OK

THE CORRECT ANSWERS ARE: B, C, F, H

30. Reaction at Joint A = 1/2 total applied load (due to symmetry)
 $= (1/2)(50 + 100 + 50) = 100 \text{ kips}$

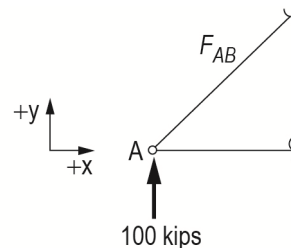
Considering Joint A:

$$\sum F_y = 0$$

$$+ \frac{3}{5} F_{AB} + 100 = 0$$

$$F_{AB} = -\frac{5}{3}(100)$$

$$= -167 \text{ kips (compression)}$$



THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

31. Reference: AISC, 15th ed., Table 3-23, Cases 26 and 27, the point of interest, $x_1 = a$.

$$\frac{wl^3a}{24EI} = \frac{Pa}{6EI}(2al + 3a^2 - a^2)$$

$$P = \frac{wl^3}{4(2al + 2a^2)}$$

$$P = \frac{(500 \text{ lb/ft})(15 \text{ ft})^3}{4[2(5 \text{ ft})(15 \text{ ft}) + 2(5 \text{ ft})^2]} = 2,109 \text{ lb} = 2.11 \text{ kip}$$

Alternatively, you could calculate the upward displacement at the free end due to the distributed load and then solve for the point load, **P**, which results in the same displacement.

THE CORRECT ANSWER IS: ≥ 2.05 to ≤ 2.15

32. Reference: AISC, 15th ed., Table 3-23, Superimposing Cases 1 and 9, noting that the concentrated loads are symmetrically placed at $a = l/3$; moment of inertia, I , from Table 1-1.

$$\Delta_{\max} = \frac{5wl^4}{384EI} + \frac{23Pl^3}{648EI}$$

$$\Delta_{\max} = \frac{5(1.2 \text{ kip/ft})(24 \text{ ft})^4}{384(29,000 \text{ ksi})(425 \text{ in}^4)} + \frac{23(2 \text{ kip})(24 \text{ ft})^3}{648(29,000 \text{ ksi})(425 \text{ in}^4)}$$

$$\Delta_{\max} = 0.7268 \text{ in.} + 0.1376 \text{ in.}$$

$$\Delta_{\max} = 0.864 \text{ in.}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

33. Reference: *PE Civil Reference Handbook*, Thermal Deformations section, and Hooke's Law relating stress and strain.

$$\delta_t = \alpha L(T - T_o)$$

$$\delta_t = (6.5 \times 10^{-6} / ^\circ\text{F})(20 \text{ ft})(200^\circ\text{F}) = 0.312 \text{ in.}$$

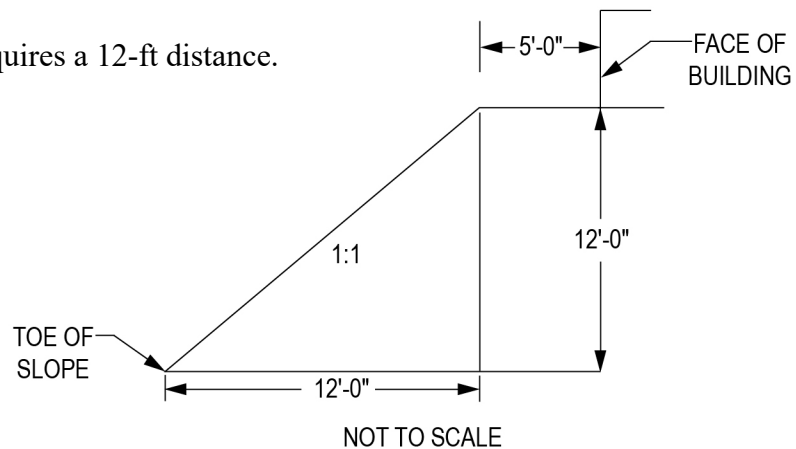
$$\sigma = E\varepsilon = E\left(\frac{\delta}{L}\right)$$

$$\sigma = 29,000 \text{ ksi} \left(\frac{0.312 \text{ in.}}{240 \text{ in.}} \right) = 37.7 \text{ ksi}$$

THE CORRECT ANSWER IS: C

34. Type B soil has a maximum permissible slope of 1:1.

Therefore, a 12-ft depth requires a 12-ft distance.



Because there is a 5-ft perimeter strip, the minimum distance from the toe of the slope to the face of the structure = 12 ft + 5 ft = 17 ft.

THE CORRECT ANSWER IS: C

35.
$$R = \left[(0.5 \text{ ft}(150 \text{ pcf}) + 50 \text{ psf} + 12 \text{ psf}) + (0.83 \text{ ft}(150 \text{ pcf}) + 50 + 12) \right] (6 \text{ ft} \times 6 \text{ ft})$$

$$= 11,646 \text{ lb}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

36. The temporary structure must be designed in compliance with applicable building codes, regulations, and safety standards. The submittal review process helps to identify and address potential safety issues or design flaws before the structure is installed, ensuring that the temporary structure is safe and fit for its intended use.

THE CORRECT ANSWER IS: A

37. IBC 2018, Section 1705.2.1 (Structural steel), states that "Special inspections and nondestructive testing of structural steel elements in buildings, structures, and portions thereof shall be in accordance with the quality assurance inspection requirements of AISC 360."

Therefore, per AISC 360-16 (*AISC Steel Construction Manual*), Chapter N (Quality Control and Quality Assurance), N5.6 (Minimum Requirements for Inspection of Structural Steel Buildings - Inspection of High-Strength Bolting) on p. 16.1-500:

Snug-tightened joints are required to be inspected to ensure the following:

1. Proper fastener components are used; therefore, proper product marking in accordance with the applicable ASTM standard
2. The faying surfaces are brought into firm contact during installation of the bolts.

Note that the magnitude of the clamping force that exists in a snug-tightened joint is not a consideration and need not be verified. ASTM A325N bolts would indicate that the bolts are a snug-tightened joint and not pre-tensioned or slip-critical joints.

THE CORRECT ANSWERS ARE: C, D

38. The angle between the brace and horizontal is $50.9^\circ = \arctan(6 \text{ ft}/5 \text{ ft})$

Therefore, the 15-kip compressive force on the diagonal brace would be a horizontal force at the top of the brace:

Horizontal force at the top of the brace = $15 \text{ kips} \times \cos(50.2^\circ) = 9.6 \text{ kips}$

Therefore, the **H** force is $6 \text{ ft}/8 \text{ ft} (9.6 \text{ kips}) = 7.2 \text{ kips}$

THE CORRECT ANSWER IS: ≥ 7.1 to ≤ 7.3

PE CIVIL: STRUCTURAL SOLUTIONS

39. According to AISC, 15th ed., Table 3-23 (p. 3-208), the maximum deflection is calculated as shown below.

$$\Delta_{\max} (\text{at center}) = \frac{5 w l^4}{384 EI}$$

Therefore, the minimum moment of inertia is

$$I = \frac{5 \left[(0.78 \text{ k/ft})(20 \text{ ft})^4 (12)^3 \right]}{(384 \times 29,000 \text{ ksi} \times 0.25 \text{ in.})} = 387.3 \text{ in}^4$$

$$(384 \times 29,000 \text{ ksi} \times 0.25 \text{ in.}) = 387.3 \text{ in}^4$$

Based on Table 1-1, the following beams have the associated moment of inertia:

Beam	I_{xx}	Acceptable? (or greater than 387.3 in ⁴)
W10×49	272	No
W12×26	204	No
W12×53	425	Yes
W14×22	199	No
W14×43	428	Yes
W16×26	301	No

THE CORRECT ANSWERS ARE: C, E

40. This is the largest radius that has at least 42,000-lb allowable capacity.

Radius in Feet	Main Boom Length in Feet									
	34.9	43.7	52.5	61.4	70.2	79	87.8	96.7	105.5	114.3
8	110,000 (68.5)	—	—	—	—	—	—	—	—	—
10	88,750 (65)	81,000 (70.5)	79,700 (74)	79,200 (77)	72,650 (79.5)	—	—	—	—	—
12	80,150 (61)	78,100 (67.5)	76,400 (72)	75,050 (75)	68,200 (77.5)	*57,800 (80)	—	—	—	—
15	69,550 (55)	68,200 (63)	66,700 (68)	65,450 (72)	60,450 (75)	51,050 (77)	42,250 (79)	*34,500 (80)	—	—
20	55,300 (43)	55,800 (55.5)	54,900 (62)	53,900 (67)	49,400 (70.5)	42,050 (73)	34,950 (75.5)	33,300 (77)	28,100 (78.5)	*25,800 (80)
25	40,400 (26)	43,300 (46.5)	43,750 (55.5)	44,100 (61.5)	41,300 (66)	35,150 (69)	29,250 (72)	28,150 (74)	27,000 (76)	24,600 (77.5)
30	—	34,750 (35)	35,200 (48)	35,500 (56)	35,250 (61.5)	29,950 (65)	24,900 (68.5)	23,950 (71)	23,150 (73)	21,800 (75)
35	—	24,950 (17)	28,150 (39.5)	28,300 (50)	28,450 (56.5)	25,950 (61)	21,500 (64.5)	20,650 (67.5)	20,000 (70)	19,400 (72)
40	—	—	22,050 (29)	22,300 (43)	22,500 (51)	22,750 (56.5)	18,750 (61)	18,050 (64.5)	17,450 (67)	16,900 (69.5)

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

41. The long-term settlement for Case I is less than Case II because clay is subject to long-term settlement.

THE CORRECT ANSWER IS: A

42. The commonly used soil classification systems for engineering applications are USCS and AASHTO. Both of these systems use gradation and Atterberg limits as two of the criteria.

THE CORRECT ANSWER IS: D

43. The SPT N-value provides an indication of the relative density of cohesionless soils.

THE CORRECT ANSWER IS: C

44. An early-strength concrete is needed with a minimum compressive strength of 3,500 psi. To achieve the requirements, a Type III cement and chemical accelerators would be necessary.

THE CORRECT ANSWER IS: D

45. Fatigue in steel is due to cyclic loading. See AISC, 15th ed., Part 16, Appendix 3.

THE CORRECT ANSWER IS: A

46. $\text{Area} = \pi d^2 / 4 = 28.3 \text{ in}^2$

Compressive stress = axial load/area

$$\text{Sample 1} \quad f'_c = \frac{65,447}{28.3} = 2,313 \text{ psi}$$

$$\text{Sample 2} \quad f'_c = \frac{63,617}{28.3} = 2,248 \text{ psi}$$

$$\text{Sample 3} \quad f'_c = \frac{69,872}{28.3} = 2,469 \text{ psi}$$

$$\text{Average} = \frac{(2,313 + 2,248 + 2,469)}{3} = 2,343 \text{ psi}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

47. Reference: *PE Civil Reference Handbook*, Weight-Volume Relationships section.

$$\text{Total density } (\gamma) = \frac{W}{V} = \frac{W_s + W_w}{V_s + V_w + V_a}$$

where γ = total density

W = total weight

V = total volume

W_s = weight soil

W_w = weight of water

V_s = volume of soil

V_w = volume of water

V_a = volume of air

$$\gamma = \frac{9.11 \text{ lb} - 4.41 \text{ lb}}{0.03 \text{ ft}^3} = 156.67 \text{ lb/ft}^3 \text{ (pcf)}$$

$$\text{Dry unit weight of soil } (\gamma_d) = \frac{\gamma}{1 + w}$$

where w = moisture content

$$\gamma_d = \frac{156.67 \text{ pcf}}{1 + 0.115} = 140.51 \text{ pcf}$$

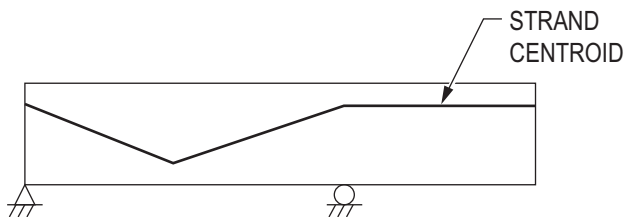
THE CORRECT ANSWER IS: B

48. ACI 318-14, Tables 19.3.1.1, 19.3.2.1, and 19.3.3.1 give 4.5%. Neglect tolerance per problem statement.

$\therefore$ minimum = 4.5%

THE CORRECT ANSWER IS: A

49.



THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

50. Reference: NDS 2018

$$F_{cE} = \frac{0.822 E'_{\min}}{(l_e/d)^2} \quad \text{Section 3.7.1}$$

$$E'_{\min} = E_{\min} \times C_M \times C_t \times C_i \times C_T \quad \text{Table 4.3.1}$$

where:

$$E_{\min} = 440,000 \text{ psi (given)}$$

$$C_M = 1.0 \text{ (dry service)} \quad \text{Section 4.3.3}$$

$$C_t = 1.0 \text{ (normal temperature)} \quad \text{Table 2.3.3}$$

$$C_i = 1.0 \text{ (given)}$$

$$C_T = 1.0 \text{ (not a truss member)} \quad \text{Section 4.3.11}$$

$$\begin{aligned} E'_{\min} &= 440,000 \text{ psi} \times 1.0 \times 1.0 \times 1.0 \times 1.0 \\ &= 440,000 \text{ psi} \end{aligned}$$

$$\begin{aligned} \text{Check } l_e/d: \quad &3 \text{ 1/2-in. direction} && 8 \times 12 / 3.5 = 27.5 \\ &1 \text{ 1/2-in. direction} && 4 \times 12 / 1.5 = 32 \quad \text{Controls} \end{aligned}$$

$$F_{cE} = \frac{0.822 \times 440,000}{32^2} = 353 \text{ psi}$$

THE CORRECT ANSWER IS: B

51. Reference: ASCE 7-16, Design Loads Table 3.2-1, p. 11.

$$85 \text{ lb/ft}^2$$

THE CORRECT ANSWER IS: C

52. Reference: ACI 318-14, Section 26.12.3.1(b)

Concrete is unsatisfactory. Every arithmetic average of any three consecutive strength tests should be equal to or exceed the required f'_c .

$$\frac{4,215 + 4,160 + 3,625}{3} = 4,000; \quad \frac{4,160 + 3,625 + 4,010}{3} = 3,931; \quad \frac{3,625 + 4,010 + 4,015}{3} = 3,883$$

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

53. Reference: AISC, 15th ed., Part 16

$$r_{EW} = r_{NS} \quad K_{EW} > K_{NS}$$

Buckling in east-west direction will control.

Table C-A-7.1

THE CORRECT ANSWER IS: B

54. Concrete reaches ultimate strain and fails prior to steel yielding. Failure is sudden without warning (brittle failure).

THE CORRECT ANSWER IS: D

55. Reference: NDS 2018

Douglas Fir-Larch No. 2, $F_c = 1,350$ psi

Table 4A Supplement

$$F_c^* = (F_c)(C_D)(C_M)(C_t)(C_F)(C_i)$$

Table 4.3.1

$$= (1,350)(1.6)(1.0)(1.0)(1.15)(1.0) = 2,484 \text{ psi}$$

$$F_c' = (F_c^*)(C_P)$$

Table 4.3.1

$$= (2,484)(0.33) = 819.72 \text{ psi}$$

$$A = bd = (1.5)(3.5) = 5.25 \text{ in}^2$$

$$P = F_c' A = (819.72)(5.25) = 4,303 \text{ lb}$$

THE CORRECT ANSWER IS: D

PE CIVIL: STRUCTURAL SOLUTIONS

56. Reference: TMS 402/602, 2016, paragraph 8.3.5.1.4

The allowable shear stress resisted by the steel reinforcement is given by:

$$F_{vs} = 0.5 \left[\frac{A_v F_s d_v}{A_{nv} s} \right] \quad \text{Equation 8-27}$$

where

0.5 is an empirical factor

Commentary Section 9.3.4.1.2.2

A_v = cross-sectional area of shear reinforcement

= 0.20 in² for #4 bars

Table CC-6.1.2

F_s = 32,000 psi for Grade 60 reinforcement

Section 8.3.3.1 (b)

A_{nv} = net cross-sectional area of member

d_v = actual depth of member = 24 in.

The allowable shear force V is given:

$$V = F_{vs} A_{nv}$$

Therefore,

$$\begin{aligned} V &= 0.5 \left[\frac{A_v F_s d_v}{s} \right] \\ &= 0.5 \left(\frac{0.2(32,000)(24)}{8(1,000)} \right) \\ &= 0.5(19.2) \\ &= 10 \text{ kips} \end{aligned}$$

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

57. Reference: ACI 318-14, Section 6.5, Table 6.5.2

Two or more spans not exceeding 10 ft.

Spans are equal.

Loads are uniformly distributed.

$LL \leq 3DL$

Prismatic members

$$l_n = 10 \text{ ft} - 1 \text{ ft} = 9 \text{ ft per Sec. 6.5}$$

$$W_u = 1.2 \times 100 + 1.6 \times 50 = 200 \text{ plf}$$

Use $W_u l_n^2 / 12$ for ALL supports.

Note: Beam/column stiffness not applicable to slabs

$$M_u = W_u l_n^2 / 12 = 1.35 \text{ ft-kips/ft}$$

THE CORRECT ANSWER IS: A

58. Reference: AISC, 15th ed.

The properties of a W10×22 are as follows:

Table 3-2 or Table 3-10.

$$\phi_b M_{px} = 97.5 \text{ ft-kips}$$

$$M_{px} / \Omega_b = 64.9 \text{ ft-kips}$$

$$BF / \Omega_b = 2.68 \text{ kips (ASD) or } \phi_b BF = 4.02 \text{ kips (LRFD)}$$

$$L_p = 4.7 \text{ ft}$$

$$L_b = 10 \text{ ft}$$

$$\text{ASD: } M_{px} / \Omega_b - BF / \Omega_b (L_b - L_p) = M_n / \Omega_b$$

Equation 3-4b

$$64.9 - 2.68(10 - 4.7) = 50.7 \text{ ft-kips}$$

$$\text{LRFD: } \phi_b M_{px} - \phi_b BF (L_b - L_p) = \phi_b M_n$$

Equation 3-4a

$$97.5 - 4.02(10 - 4.7) = 76.2 \text{ ft-kips}$$

THE CORRECT ANSWER IS: B

59. $A' = \text{area of flange} = 6 \text{ in.}(0.5 \text{ in.}) = 3 \text{ in}^2$

$\bar{y}' = \text{distance from neutral axis to centroid of flange} = 5.75 \text{ in.}$

$$Q_{\text{flange}} = 3 \text{ in}^2 (5.75 \text{ in.}) = 17.25 \text{ in}^3$$

$$q = \frac{VQ}{I} = \frac{20 \text{ kips}(17.25 \text{ in}^3)}{232 \text{ in}^4} = 1.5 \text{ kips/in.}$$

THE CORRECT ANSWER IS: A

PE CIVIL: STRUCTURAL SOLUTIONS

60.
$$\frac{(1.6 \text{ kips})(3 \text{ in.})}{(8.25 \text{ in.} - 0.50 \text{ in.})} = 0.619 \text{ kips} = 619 \text{ lb}$$

THE CORRECT ANSWER IS: B

61. Reference: AISC, 15th ed.

By inspection P controls.

$$M_n = M_p = F_y Z_x \leq 1.6 M_y$$

$$F_y = 36 \text{ ksi}$$

$$S_x = 1/6 b d^2 = 1/6 (8)(0.375)^2 = 0.1875 \text{ in}^3$$

$$M_y = F_y S_x = (36)(0.1875) = 6.75 \text{ in.-kips}$$

$$Z_x = 1/4 b d^2 = 1/4 (8)(0.375)^2 = 0.2813 \text{ in}^3$$

$$M_p = F_y Z_x = (36)(0.2813) = 10.125 \text{ in.-kips} \quad \leftarrow \text{governs}$$

$$\text{Check } M_p \leq 1.6 M_y \quad 1.6(6.75) = 10.8 > 10.125 \therefore \text{OK}$$

AISC ASD:

$$M_n / \Omega = 10.125 / 1.67 = 6.06 \text{ in.-kips}$$

$$P_{\text{allow}} = \frac{6.06}{\left[4 - (3/8)\right]} = 1.67 \sim 1.7 \text{ kips}$$

AISC LRFD:

$$\phi M_n = 0.9(10.125) = 9.11 \text{ in.-kips}$$

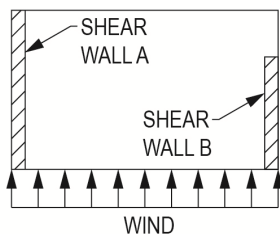
$$\phi P_n = \frac{9.11}{\left[4 - (3/8)\right]} = 2.51 \sim 2.5 \text{ kips}$$

THE CORRECT ANSWER IS: C

PE CIVIL: STRUCTURAL SOLUTIONS

62. Reference: ASCE 7-16, Section 12.3.

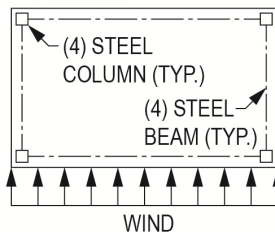
- WOOD
- DIAPHRAGM DEFLECTION CAN BE TOLERATED



Flexible

Torsionally irregular

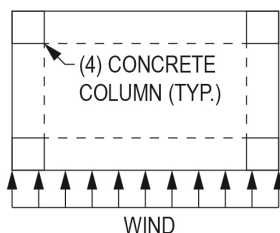
- METAL DECK SUPPORTED BY STEEL BEAMS AND COLUMNS



Flexible

Torsionally regular

- PT CONCRETE SLAB SUPPORTED BY CONCRETE COLUMNS/BEAMS

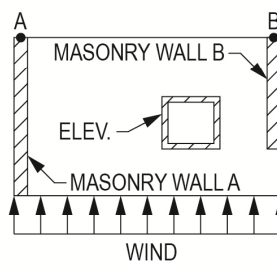


Rigid

Torsionally regular

- CONCRETE SLAB SUPPORTED BY MASONRY WALLS

$$A_{\text{STORYDRIFT}} = 1.5 \left(\frac{A_{\text{STORYDRIFT}} + B_{\text{STORYDRIFT}}}{2} \right)$$



Torsionally irregular

Rigid

THE CORRECTLY LABELED DIAGRAMS ARE SHOWN ABOVE.

PE CIVIL: STRUCTURAL SOLUTIONS

63. $d \approx 24 \text{ in.} - (3 \text{ in.})_{\text{cov}} - (1 \text{ in.})_{\#8} = 20 \text{ in.}$ (given) Use d at $\mathcal{C}_L$ of # 8 layers

b_o at $d/2$ from face (critical section)

$$b_o = 4(32 \text{ in.}) = 128 \text{ in.}$$

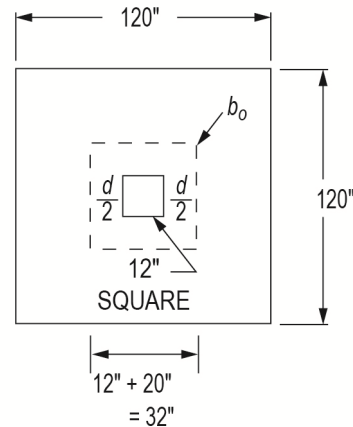
Per ACI 318-14, Section 22.6.5.2, Table 22.6.5.2, Section 22.6.5.3,
use smallest of the following:

a. $\left(2 + \frac{4}{1/1}\right) = 6$

b. $\left[\frac{(40)(20 \text{ in.})}{128 \text{ in.}} + 2\right] = 8.3$

c. 4 Controls

$$\begin{aligned}\phi V_n &= 0.75(4)(1)(\sqrt{3,000})(128 \text{ in.})(20 \text{ in.}) \\ &= 420.6 \text{ kips}\end{aligned}$$



THE CORRECT ANSWER IS: D

64. Critical case is skin friction resistance = applied uplift load.

$$\frac{(20 \text{ ft})\pi D(200 \text{ psf})}{1,000} = 100 \text{ kips}$$

$$D = \frac{100 \text{ kips} \times 1,000}{(200 \text{ psf} \times 20 \text{ ft} \times \pi)}$$

$$D = 7.96 \text{ ft}$$

THE CORRECT ANSWER IS: A

65. Reference: AISC, 15th ed.

$$R_n = F_n A_b$$

Equation J3-1

$$\phi = 0.75, \Omega = 2.00$$

ASD:

$$F_{nv}/\Omega = 13.5 \text{ ksi} \quad F_{nv} = 27 \text{ ksi}$$

Table J3.2

$$\text{Allowable load} = 2(13)(0.79) = 21.2 \text{ kips}$$

LRFD:

$$\phi R_n = \phi F_{nv} A_b$$

$$\phi F_{nv} = 20.3 \text{ ksi (A307 bolts)}$$

$$\phi R_n = (20.3)(0.79)(2) = 31.9 \text{ kips}$$

Alternate solution, use Table 7-1

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

- 66.** Vertical reaction to Footing $B = 150 \text{ kips} \times 5 \text{ ft}/6 \text{ ft} = 125 \text{ kips}$
 Footing spring constant $K = 100 \text{ lb/in}^3 \times 24 \text{ in.} \times 24 \text{ in.} = 57,600 \text{ lb/in.}$
 $F = K\Delta$
 $\Delta = F/K = 125 \times 1,000/57,600 = 2.2 \text{ in.}$

THE CORRECT ANSWER IS: A

- 67.** Reference: PCI, 7th ed., Chapter 4.

Total prestress force, $P = 2 \times 6 \text{ strands} \times 0.153 \text{ in}^2/\text{strand} \times 175 \text{ ksi} = 321.3 \text{ kips}$
 Axial stress $= P/A = 321.3/(36 \times 18) = 0.496 \text{ ksi}$ (compression)
 Bending stress due to prestress $= Pe/S$ $S = bd^2/6 = 18 \times 36^2/6 = 3,888 \text{ in}^3$
 $e = 36/2 - 3 - 2/2 = 14 \text{ in.}$
 Bending stress due to prestress $= 321.3 \times 14/3,888 = 1.158 \text{ ksi}$ (tension at top)
 Bending stress due to beam self-weight $= 0.65 \text{ ksi}$ (compression at top)
 Thus, stress at top at release $= 0.49 - 1.15 + 0.65 = -0.01 \text{ ksi}$ (tension)

THE CORRECT ANSWER IS: A

- 68.** Reference: AASHTO, 8th ed., Section 9.7.2.3

$L_{\text{eff}} = 8.5 \text{ ft} - b_f + \text{flange overhang}$
 $= 8.5 \text{ ft} - 1.5 \text{ ft} + 0.25 \text{ ft} + 0.25 \text{ ft}$
 $= 7.5 \text{ ft}$ (7 ft 6 in.)

THE CORRECT ANSWER IS: C

- 69.** Reference: IBC 2018

The presumptive soil values are allowable stress values.

The presumptive vertical foundation pressure for a sandy gravel and gravel (GW and GP) material is 3,000 psf (3 ksf). (Table 180 6.2)

Using the stated foundation plan size of 8 ft $\times$ 8 ft, the allowable vertical load in kips is $3 \text{ (ksf)} \times 8 \text{ ft} \times 8 \text{ ft} = 192 \text{ k.}$

THE CORRECT ANSWER IS: ≥ 190 to ≤ 194

PE CIVIL: STRUCTURAL SOLUTIONS

70. Reference: IBC 2018, Section 1810.3.3.1.5

1810.3.3.1.5 Uplift capacity of a single deep foundation element. Where required by the design, the uplift capacity of a single deep foundation element shall be determined by an approved method of analysis based on a minimum factor of safety of three or by load tests conducted in accordance with ASTM D3689. The maximum allowable uplift load shall not exceed the ultimate load capacity as determined in Section 1810.3.3.1.2, using the results of load tests conducted in accordance with ASTM D3689, divided by a factor of safety of two.

Exception: Where uplift is due to wind or seismic loading, the minimum factor of safety shall be two where capacity is determined by an analysis and one and one-half where capacity is determined by load tests.

THE CORRECT ANSWERS ARE: A, C

71. Reference: IBC 2018, Section 1810.4.1.1

1810.4.1.1 Compressive strength of precast concrete piles. A precast concrete pile shall not be driven before the concrete has attained a compressive strength of not less than 75% of the specified compressive strength (f'_c), but not less than the strength sufficient to withstand handling and driving forces.

THE CORRECT ANSWERS ARE: A, C

PE CIVIL: STRUCTURAL SOLUTIONS

72. Reference: AISC, 15th ed.

D2. TENSILE STRENGTH

The design tensile strength, $\phi_t P_n$, and the allowable tensile strength, P_n/Ω_t , of tension members shall be the lower value obtained according to the limit states of tensile yielding in the gross section and tensile rupture in the net section.

(a) For tensile yielding in the gross section

$$P_n = F_y A_g \quad (\text{D2-1})$$

$$\phi_t = 0.90 \text{ (LRFD)} \quad \Omega_t = 1.67 \text{ (ASD)}$$

(b) For tensile rupture in the net section

$$P_n = F_u A_e \quad (\text{D2-2})$$

$$\phi_t = 0.75 \text{ (LRFD)} \quad \Omega_t = 2.00 \text{ (ASD)}$$

E1. GENERAL PROVISIONS

The design compressive strength, $\phi_c P_n$, and the allowable compressive strength, P_n/Ω_c , are determined as follows.

The nominal compressive strength, P_n , shall be the lowest value obtained based on the applicable limit states of flexural buckling, torsional buckling, and flexural-torsional buckling.

$$\phi_c = 0.90 \text{ (LRFD)} \quad \Omega_c = 1.67 \text{ (ASD)}$$

F1. GENERAL PROVISIONS

The design flexural strength, $\phi_b M_n$, and the allowable flexural strength, M_n/Ω_b , shall be determined as follows:

(a) For all provisions in this chapter

$$\phi_b = 0.90 \text{ (LRFD)} \quad \Omega_b = 1.67 \text{ (ASD)}$$

G1. GENERAL PROVISIONS

The design shear strength, $\phi_v V_n$, and the allowable shear strength, V_n/Ω_v , shall be determined as follows:

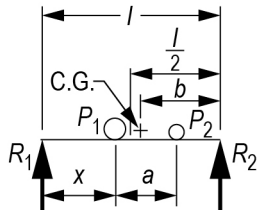
(a) For all provisions in this chapter except Section G2.1(a)

$$\phi_v = 0.90 \text{ (LRFD)} \quad \Omega_v = 1.67 \text{ (ASD)}$$

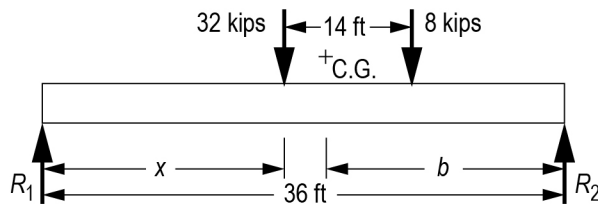
THE CORRECT ANSWERS ARE: A, C, D, E

PE CIVIL: STRUCTURAL SOLUTIONS

73. Reference: AISC, 15th ed., p. 3-223, provides instructions for calculating maximum moment for simple beams carrying moving concentrated loads.



Maximum moment occurs under P_1 when $x = b$.



$$x = 36 \text{ ft} - x - 2.8 \text{ ft}$$

$$2x = 33.2 \text{ ft} \quad \therefore x = 16.6 \text{ ft}$$

Calculate the reactions:

$$\Sigma M = 0 = 32 \text{ kips} (16.6 \text{ ft}) + 8 \text{ kips} (30.6 \text{ ft}) - R_2 (36 \text{ ft}) \quad \therefore R_2 = 21.6 \text{ kip}$$

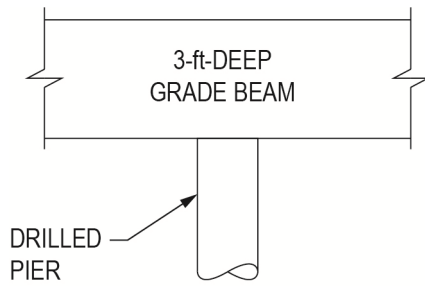
$$\Sigma F_y = 0 = R_1 - 32 \text{ kips} - 8 \text{ kips} + 21.6 \text{ kips}, \quad \therefore R_1 = 18.4 \text{ kips}$$

Maximum moment occurs under P_1 $\therefore M_{\max} = 18.4 \text{ kips} (16.6 \text{ ft}) = 305 \text{ ft-kips}$

THE CORRECT ANSWERS ARE: ≥ 300 to ≤ 310

PE CIVIL: STRUCTURAL SOLUTIONS

74. Reference: ASCE 7-16, Section 12.1.4, p. 89



Factored reactions from grade beam to drilled pier:

LL = 10 kips

DL = 8 kips

The minimum design strength is 5% of DL + LL reactions:

$0.05 (18 \text{ kips}) = 900 \text{ lb}$

THE CORRECT ANSWER IS: C

75. Reference: AISC, 15th ed., Chapter J, Section 3

J 3.3 min spacing = $2 \frac{2}{3} (d)$

$2 \frac{2}{3} (5/8) = 1.67 \text{ in.}$

J 3.4 min edge distance $5/8\text{-in. bolt} = 7/8 \text{ in.} + 1/8 \text{ in. for short slot} = 1 \text{ in.}$

J 3.5 max edge distance $12(t) = 12(3/4) = 9 \text{ in. or } 6 \text{ in.}$

THE CORRECT ANSWERS ARE: B, C, F

76. Reference: ASCE 7-16, Section 12.1.3, p. 89

The connection design strength shall be capable of transmitting a seismic force of whichever is greater:

$0.133 S_{DS} \times \text{weight}$ or 5% of smaller portion's weight

$0.133 (0.439)(70,000) = 4,087 \text{ lb}$

$0.05 (70,000) = 3,500 \text{ lb}$

$\sim 4,100 \text{ lb}$

THE CORRECT ANSWER IS: B

PE CIVIL: STRUCTURAL SOLUTIONS

77. Reference: ACI 318-14, Section 17.3.3

Strength reduction factor is 0.55

Anchor governed by concrete pullout = 17.3.3c

Medium reliability = Category 2

Supplemental reinforcement is not present = Condition B

Tension loads but superseded by above factors

THE CORRECT ANSWER IS: D

78. Reference: ACI 318-14, Section 17.4.1.2

$$N_{sa} = A_{se,N} \times f_{uta}$$

$$A_{se,N} \text{ for a \#6} = 0.44 \text{ in}^2$$

$$f_{uta} = \text{smaller of } 1.9 f_{ya} \text{ and } 125,000 \text{ psi}$$

$$f_{ya} = 60 \text{ ksi (1.9)} = 114,000 \text{ psi, which is less than } 125,000 \text{ psi}$$

$$N = 0.44 \text{ in}^2 (114,000 \text{ psi}) = 50,160 \text{ lb}$$

THE CORRECT ANSWER IS: 50,160

PE CIVIL: STRUCTURAL SOLUTIONS

79. Reference: *PE Civil Reference Handbook*

$$\begin{aligned}
 FS_s &= \frac{(W + P_v) \tan \delta_b + C_a B}{P_h} \\
 &= \frac{4,200 \text{ lb/ft} (\tan 30^\circ) + 500 \text{ psf} (4 \text{ ft})}{1,623 \text{ lb/ft}} \\
 &= 2.73
 \end{aligned}$$

$$P_v = 0$$

$$\begin{aligned}
 W &= \frac{1}{2} (3 \text{ ft} + 4 \text{ ft}) (8 \text{ ft}) (150 \text{ pcf}) \\
 &= 4,200 \text{ lb/ft}
 \end{aligned}$$

$$\begin{aligned}
 P_h &= \frac{K_a \times z^2}{2} \\
 &= \frac{0.40586 (125 \text{ psf}) (8 \text{ ft})^2}{2} \\
 &= 1,623 \text{ lb/ft}
 \end{aligned}$$

$$\begin{aligned}
 K_a &= \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) \\
 &= \tan^2 \left(45^\circ - \frac{25^\circ}{2} \right) \\
 &= 0.40586
 \end{aligned}$$

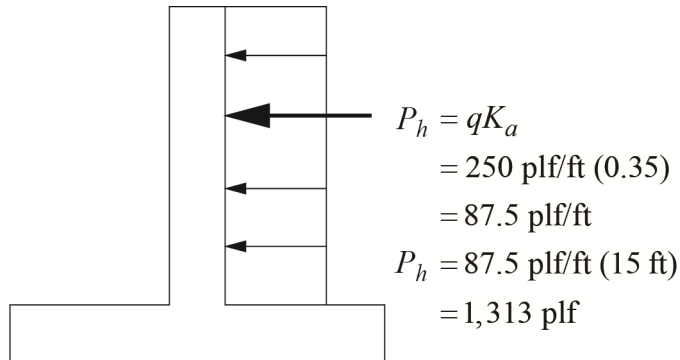
THE CORRECT ANSWER IS: ≥ 2.70 to ≤ 2.75

PE CIVIL: STRUCTURAL SOLUTIONS

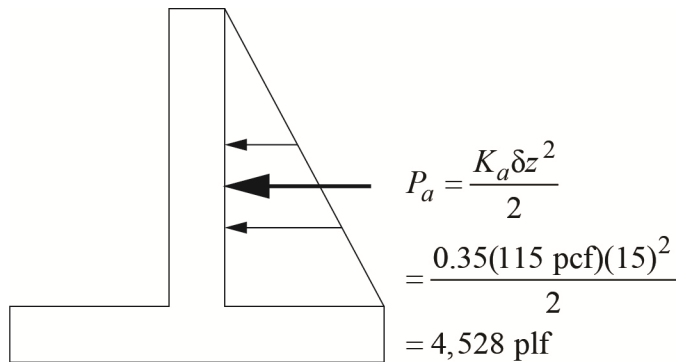
80. Reference: *PE Civil Reference Handbook*

Active earth pressure due to:

SURCHARGE



BACKFILL



Total force/unit width = 1,313 plf + 4,528 plf
= 5,841 plf

THE CORRECT ANSWER IS: A